

Electrodynamics

Covariant Formulation of Electrodynamics

$$\vec{\nabla} \cdot \vec{E} = 4\pi \rho^{\text{charge density}}, \quad \vec{\nabla} \times \vec{B} - \frac{1}{c} \frac{\partial \vec{E}}{\partial t} = \frac{4\pi}{c} \vec{J}^{\text{current density}}$$

$$\vec{\nabla} \cdot \vec{B} = 0, \quad \vec{\nabla} \times \vec{E} + \frac{1}{c} \frac{\partial \vec{B}}{\partial t} = 0$$

1) Continuity eqn: $\frac{1}{c} \frac{\partial}{\partial t} (\vec{\nabla} \cdot \vec{E}) = \frac{4\pi}{c} \frac{\partial \rho}{\partial t}$

assume $\frac{1}{c} \frac{\partial}{\partial t} \vec{\nabla} \cdot \vec{E} = \vec{\nabla} \cdot \frac{\partial \vec{E}}{\partial t}$

$$\underbrace{\vec{\nabla} \cdot \left\{ \vec{\nabla} \times \vec{B} - \frac{1}{c} \frac{\partial \vec{E}}{\partial t} \right\}}_{=0} = \frac{4\pi}{c} \vec{\nabla} \cdot \vec{J}$$

$$\rightarrow \frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot \vec{J} = 0$$

Electrodynamics 23/01

$$J^\mu = (\rho, \vec{j}) \quad \text{continuity eq } \partial_\mu J^\mu = 0$$

$$A^\mu = (\phi, \vec{A})$$

Gauge freedom $\left\{ \begin{array}{l} \vec{A} \rightarrow \vec{A} + \vec{\nabla} \chi \\ \phi \rightarrow \phi - \frac{1}{c} \frac{\partial \chi}{\partial t} \end{array} \right\} \rightarrow \vec{E}, \vec{B} \text{ unchanged}$
 $F^{\mu\nu}$ also gauge invariant

$$A^\mu \rightarrow A^\mu + \partial^\mu \chi$$

Maxwells in terms of A, ϕ

$$\nabla^2 \phi + \frac{1}{c} \frac{\partial}{\partial t} (\vec{\nabla} \cdot \vec{A}) = -\frac{4\pi}{c} \rho$$

$$\nabla^2 \vec{A} - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \vec{A} - \underbrace{\vec{\nabla} \left(\vec{\nabla} \cdot \vec{A} + \frac{1}{c} \frac{\partial \phi}{\partial t} \right)}_{\text{Problem}} = -\frac{4\pi}{c} \vec{j}$$

Lorentz gauge:

$$\text{find } \chi \text{ s.t. } \vec{\nabla} \cdot \vec{A}' + \frac{1}{c} \frac{\partial \phi'}{\partial t} = 0$$

$$\left. \begin{array}{l} \nabla^2 \phi - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \phi = -\frac{4\pi}{c} \rho \\ \vec{\nabla} \cdot \vec{A}' - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \vec{A}' = -\frac{4\pi}{c} \vec{j} \end{array} \right\} \begin{array}{l} \overbrace{\left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right)}^{=-\square} A^\mu = -\frac{4\pi}{c} J^\mu \\ \square A^\mu = \frac{4\pi}{c} J^\mu \quad \partial_\mu A^\mu = 0 \end{array}$$

$$F^{\alpha\beta} = \begin{bmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & -B_z & B_y \\ E_y & B_z & 0 & B_x \\ E_z & -B_y & B_x & 0 \end{bmatrix}$$

$$F_{\alpha\beta} = g_{\alpha\gamma} g_{\beta\delta} F^{\gamma\delta} = F^{\alpha\beta} \Big|_{\vec{E} \rightarrow -\vec{E}}$$

$$g_{\mu\nu} = \begin{bmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix}$$

dual tensor: $\tilde{F}^{\alpha\beta} = \frac{1}{2} \epsilon^{\alpha\beta\gamma\delta} F_{\gamma\delta} = \begin{bmatrix} 0 & -B_x & -B_y & -B_z \\ B_x & 0 & E_z & -E_y \\ B_y & -E_z & 0 & E_x \\ B_z & E_y & -E_x & 0 \end{bmatrix}$

$$\partial_\alpha \partial^\alpha F^{\beta\gamma} + \partial_\alpha \partial^\beta F^{\gamma\alpha} + \partial_\alpha \partial^\gamma F^{\alpha\beta} = 0$$

$$\square F^{\beta\gamma} + \partial^\beta (-\partial_\alpha F^{\alpha\gamma}) + \partial^\gamma (\partial_\alpha F^{\alpha\beta}) = 0$$

$$\square F^{\beta\gamma} = \frac{4\pi}{c} (\partial^\beta j^\gamma - j^\beta \partial^\gamma)$$

Electrodynamics

29/01

Time Independent Maxwell Eqs

$$\partial_\mu F^{\mu\nu} = \frac{4\pi}{c} j^\nu, \quad \partial_\mu F^{\mu\nu} = 0$$

$$\partial^\nu F^{\mu\rho} + \partial^\mu F^{\rho\nu} + \partial^\rho F^{\nu\mu} = 0$$

$\Rightarrow$ wave eqn

$$\underbrace{\partial_\mu \partial^\mu}_{\square} F^{\rho\sigma} = \frac{4\pi}{c} (\partial^\rho j^\sigma - \partial^\sigma j^\rho) \rightarrow \begin{cases} \square \vec{E} = -4\pi (\vec{\nabla} \rho + \frac{1}{c^2} \frac{\partial \vec{j}}{\partial t}) \\ \square \vec{B} = \frac{4\pi}{c} \vec{\nabla} \times \vec{j} \end{cases}$$

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$$

||⁰ for Lorentz gauge

$$\frac{4\pi}{c} j^\nu = \partial_\mu F^{\mu\nu} = \partial_\mu \partial^\mu A^\nu - \partial^\nu (\partial_\mu A^\mu)$$

Lorentz gauge $\square A^\nu = \frac{4\pi}{c} j^\nu \rightarrow \square \phi = 4\pi \rho$
 $\square \vec{A} = \frac{4\pi}{c} \vec{j}$

problem

start with Arbitrary A^μ convert to Lorentz Gauge

$$\text{find } \chi(x) \text{ s.t. } A'^\mu = A^\mu + \partial^\mu \chi$$

$$\text{Lorentz gauge condition: } 0 = \partial_\mu A'^\mu = \partial_\mu (A^\mu + \partial^\mu \chi) = \partial_\mu A^\mu + \partial_\mu \partial^\mu \chi$$

$$\rightarrow \square \chi = -\partial_\mu A^\mu$$

$$\text{Wave Eqn: } \square \psi(\vec{x}, t) = \left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \vec{\nabla}^2 \right) \psi(\vec{x}, t) = 4\pi f(\vec{x}, t)$$

special soln of homogeneous eq. $\square \psi = 0$

$$\psi(\vec{x}, t) = g(t - \frac{1}{c} \vec{n} \cdot \vec{x}) + h(t + \frac{1}{c} \vec{n} \cdot \vec{x}), \quad \vec{n}^2 = 1$$

g, h smooth functions

$$\frac{1}{c^2} \frac{\partial^2}{\partial t^2} g(t - \frac{1}{c} \vec{n} \cdot \vec{x}) = \frac{1}{c^2} g''$$

$$\vec{\nabla}^2 g = \sum_{k=1}^3 \frac{\partial^2}{\partial x_k^2} g = \sum_{k=1}^3 g'' \left(\delta_{kk} - \frac{1}{c} n_k \right) \left(-\frac{1}{c} n_k \right) = g'' \frac{1}{c^2} \vec{n}^2 = \frac{1}{c^2} g'' \Rightarrow \square \psi = 0$$

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Wave Equation

interpretation

$g(t - \frac{\vec{n} \cdot \vec{x}}{c})$ is const if its argument does not change

$$t - \frac{\vec{n} \cdot \vec{x}}{c} = t + dt - \frac{\vec{n}}{c} (\vec{x} + d\vec{x})$$

$$0 = dt - \frac{\vec{n}}{c} \cdot d\vec{x} \quad \text{or} \quad \vec{n} \frac{d\vec{x}}{dt} = c$$

special case $\Psi(\vec{x}, t) = e^{-i(\omega t - \vec{k} \cdot \vec{x})} = \cos(\omega t - \vec{k} \cdot \vec{x}) - i \sin(\omega t - \vec{k} \cdot \vec{x})$

$$\text{Re } \Psi(x, t) = \cos(\omega t - \vec{k} \cdot \vec{x})$$

$$\text{Re}(\square \Psi) = \square(\text{Re}(\Psi))$$

~~substituting~~ $\square \Psi(\vec{x}, t) \equiv (\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \vec{\nabla}^2) \Psi(\vec{x}, t) = 4\pi f(\vec{x}, t)$

$$\Psi(\vec{x}, t) = e^{-i(\omega t - \vec{k} \cdot \vec{x})}$$

$$0 = \square \Psi(\vec{x}, 0) = \left[\frac{1}{c^2} (-i\omega)^2 - \sum_{i=1}^3 (ik_i)^2 \right] e^{-i(\omega t - \vec{k} \cdot \vec{x})} = \underbrace{\left(-\frac{\omega^2}{c^2} + \vec{k}^2 \right)}_{=0} e^{-i(\omega t - \vec{k} \cdot \vec{x})}$$

$$\rightarrow \omega^2 = c^2 \vec{k}^2$$

$$\vec{n} = \frac{\vec{k}}{\omega}$$

Cauchy Problem

The wave eqn is 2nd order in the time derivative $\Rightarrow$ expect that the solution is determined for all time

Ansatz $\Psi(\vec{x}, t) = A(t) e^{i\vec{k} \cdot \vec{x}}$

$$0 = \square \Psi = \underbrace{\left(\frac{1}{c^2} \ddot{A}(t) + \vec{k}^2 A(t) \right)}_{=0} e^{i\vec{k} \cdot \vec{x}}$$

if $\Psi|_{t=0}$ and $\frac{d}{dt} \Psi|_{t=0}$ are given I.C.s

general soln $A(t) = A(0) \cos(c|\vec{k}|t) + \frac{\dot{A}(0)}{c|\vec{k}|} \sin(c|\vec{k}|t)$ no even

define $v_0(x) = \Psi(\vec{x}, 0)$, $v_0(\vec{x}) = \frac{\partial \Psi}{\partial t}(\vec{x}, 0)|_{t=0} = \int \frac{d^3 k}{(2\pi)^3} e^{i\vec{k} \cdot \vec{x}} \tilde{v}_0(\vec{k})$

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Wave Equation

$$\psi(\vec{x}, t) = \int \frac{d^3 k}{(2\pi)^3} \left(\tilde{V}_0(\vec{k}) \cos(c|\vec{k}|t) + \frac{V_0(\vec{k})}{c \cdot |\vec{k}|} \sin(c|\vec{k}|t) \right) e^{i\vec{k} \cdot \vec{x}}$$

Superposition of terms $A e^{i\vec{k} \cdot \vec{x}}$

$$\psi(\vec{x}, t) = \int d^3 y \left[\frac{\partial}{\partial t} D(\vec{x} - \vec{y}, t) U_0(\vec{y}) + D(\vec{x} - \vec{y}, t) V_0(\vec{y}) \right]$$

$$D(\vec{x}, t) = \int \frac{d^3 k}{(2\pi)^3} \frac{\sin(c|\vec{k}|t)}{c|\vec{k}|} e^{i\vec{k} \cdot \vec{x}} = \frac{1}{4\pi c|\vec{x}|} \frac{1}{c} \left\{ \delta\left(t - \frac{|\vec{x}|}{c}\right) - \delta\left(t + \frac{|\vec{x}|}{c}\right) \right\}$$

Conclusion: $\Psi(x, t)$ is determined in terms of $U_0(\vec{x}) = \Psi(\vec{x}, 0)$
and $V_0(x) = \frac{\partial}{\partial t} \Psi(x, t)|_{t=0}$

Apply to $\vec{E}, \vec{B}$

$$\rightarrow \vec{E}(\vec{x}, t) = \int d^3 y \left[\frac{\partial D(\vec{x} - \vec{y}, t)}{\partial t} \vec{E}(\vec{y}, 0) + D(\vec{x} - \vec{y}, t) \frac{\partial \vec{E}(\vec{y}, 0)}{\partial t} \right]$$

analogously for the $\vec{B}$ field

I.C.s must satisfy the maxwell eqns

$$\begin{cases} \vec{\nabla} \cdot \vec{E} = 0 = \vec{\nabla} \cdot \vec{B} \\ \vec{\nabla} \times \vec{E} + \frac{1}{c} \frac{\partial \vec{B}}{\partial t} = 0 \quad \text{at } t=0 \\ \vec{\nabla} \times \vec{B} - \frac{1}{c} \frac{\partial \vec{E}}{\partial t} = 0 \end{cases}$$

Compute $\vec{\nabla} \cdot \vec{E}(\vec{x}, t)$

trick: $\frac{\partial}{\partial x_k} D(\vec{x} - \vec{y}, t) = - \frac{\partial}{\partial y_k} D(\vec{x} - \vec{y}, t)$ and use Integration by parts w.r.t y integral

$$\begin{aligned} \vec{\nabla}_x \cdot \vec{E}(\vec{x}, t) &= \int d^3 y \left[\left(- \vec{\nabla}_y \cdot \frac{\partial}{\partial t} D(\vec{x} - \vec{y}, t) \right) \cdot \vec{E}(\vec{y}, 0) + \left(- \vec{\nabla}_y \cdot D(\vec{x} - \vec{y}, t) \right) \frac{\partial \vec{E}}{\partial t} \bigg|_{\vec{y}, 0} \right] \\ &\stackrel{I.B.P.}{=} \int d^3 y \left[\frac{\partial}{\partial t} D(\vec{x} - \vec{y}, t) \underbrace{\vec{\nabla}_y \cdot \vec{E}(\vec{y}, 0)}_{=0} + D(\vec{x} - \vec{y}, t) \cdot \underbrace{\vec{\nabla}_y \cdot \left(\frac{\partial \vec{E}}{\partial t} \right) (\vec{y}, 0)}_{= \vec{E} = c \vec{\nabla} \times \vec{B}} \right] \\ &= 0 \end{aligned}$$

≈ 0

Electrodynamics 30/01

Wave Equations

plane waves in complex notation

$$\vec{E}(\vec{x}, t) = \vec{E}_0 e^{i(\omega t - \vec{k} \cdot \vec{x})} \quad \vec{B}(\vec{x}, t) = \vec{B}_0 e^{i(\omega t - \vec{k} \cdot \vec{x})}$$

↑
const. in general
complex

$$0 = \vec{\nabla} \cdot \vec{E}(\vec{x}, t) = \vec{E}_0 \cdot (-i\vec{k}) e^{i(\omega t - \vec{k} \cdot \vec{x})} = -i\vec{k} \cdot \vec{E}(\vec{x}, t) \quad \text{same for } \vec{B}$$

$$\Rightarrow \vec{k} \cdot \vec{E} = 0 = \vec{k} \cdot \vec{B} \quad \vec{k} \text{ points in direction of wave propagation}$$

$$\vec{\nabla} \times \vec{E} + \frac{1}{c} \frac{\partial \vec{B}}{\partial t} = 0$$

insert plane wave ansatz

$$i\vec{k} \times \vec{E}_0 + \frac{1}{c} (-i\omega) \vec{B}_0 = 0$$

$$\underbrace{\vec{E}_0 \cdot \vec{k} \times \vec{E}_0}_0 = \frac{\omega}{c} \vec{B}_0 \cdot \vec{E}_0 \Rightarrow \vec{E} \perp \vec{B}$$

How to solve inhomogeneous wave eqns? ~~Use~~ Green's Functions!

Recall from Electrostatics:

$$\begin{cases} \square \phi = 4\pi \rho \\ \square \vec{A} = \frac{4\pi}{c} \vec{J} \end{cases} \xrightarrow{\text{time indep}} \begin{cases} -\nabla^2 \phi(\vec{x}) = 4\pi \rho(\vec{x}) \\ -\nabla^2 A_i(\vec{x}) = 4\pi J_i(\vec{x}) \end{cases}$$

solutions

$$\begin{aligned} \phi(\vec{x}) &= \int \frac{\rho(\vec{x}')}{|\vec{x} - \vec{x}'|} d^3x' \\ A(\vec{x}) &= \frac{1}{c} \int \frac{\vec{J}(\vec{x}')}{|\vec{x} - \vec{x}'|} d^3x' \end{aligned} \quad \begin{aligned} -\nabla_x^2 \phi(x) &= \int \rho(x') \left[\underbrace{(-\nabla_x^2) \frac{1}{|\vec{x} - \vec{x}'|}}_{4\pi \delta(\vec{x} - \vec{x}')} \right] d^3x' \\ &= 4\pi \rho(\vec{x}) \end{aligned}$$

$$-\nabla^2 \frac{1}{4\pi|\vec{x}|} = \delta^{(3)}(\vec{x})$$

Electrodynamics 31/01

Green's function for d'Alembert operator $\square$

Electrostatics: $G(\vec{x}, \vec{x}') = \frac{1}{4\pi|\vec{x}-\vec{x}'|}$, $-\nabla^2 G(\vec{x}, \vec{x}') = \delta^{(3)}(\vec{x}-\vec{x}')$

Poisson eqn: $-\nabla^2 \phi(x) = 4\pi\rho(x) \Rightarrow \phi(x) = \int d^3x' G(\vec{x}, \vec{x}') 4\pi\rho(\vec{x}') + X(x)$

Where $X(\vec{x})$ satisfies the homogeneous eq: $-\nabla^2 X(\vec{x}) = 0$

Choice of Notation: $G(\vec{x}, \vec{x}') \rightarrow G(\vec{x}-\vec{x}')$, $G(x) = \frac{1}{4\pi|\vec{x}|}$

we are looking for $D_0(x, x') \equiv D_0(x-x')$ s.t

$$\square D_0(x, x') = \delta^{(4)}(x-x')$$

or $\square D_0(x) = \delta^{(4)}(x) \rightarrow$ obtaining a soln of wave-eqn $\square \Psi = 4\pi f$

$$\Psi(x) = \int D_0(x-x') 4\pi f(x') d^4x' + X(x), \text{ with } \square X = 0$$

Compute $D_0(x)$ $D_0(x) = \int \frac{d^4k}{(2\pi)^4} e^{-ikx} \tilde{D}_0(k)$ $k \cdot x = k_\mu x^\mu = k_0 x_0 - \vec{k} \cdot \vec{x}$

inverse $\tilde{D}_0(k) = \int d^4x e^{ikx} D_0(x)$

$$\begin{aligned} D_0(x) &= \int \frac{d^4k}{(2\pi)^4} e^{-ikx} \left(\int d^4x' e^{ikx'} D_0(x') \right) \\ &= \int d^4x' \underbrace{\left(\int \frac{d^4k}{(2\pi)^4} e^{-ik(x-x')} \right)}_{\delta^{(4)}(x-x')} D_0(x') \stackrel{①}{=} D_0(x) \end{aligned}$$

$\square D_0(x) = ? \rightarrow \square e^{-ikx} = \partial_\mu \partial^\mu e^{-ikx} = \partial_\mu (-ik^\mu) e^{-ikx}$
 $= (-ik_\mu)(-ik^\mu) e^{-ikx}$
 $= -k^2 e^{-ikx}$

Electrodynamics 31/01

$$\square D_0(x) = \int \frac{d^4 k}{(2\pi)^4} e^{-ikx} (-k^2) \tilde{D}_0(k) = \delta^{(4)}(x) = \int \frac{d^4 k}{(2\pi)^4} e^{-ikx}$$

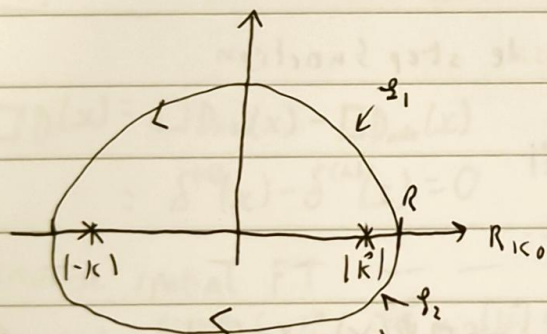
$$\Rightarrow \boxed{-k^2 \tilde{D}_0(k) = 1}$$

$$\boxed{\tilde{D}_0(k) = \frac{-1}{k^2}}$$

$$-k^2 = -k_0^2 + k^2$$

$$D_0(x) = - \int_R \frac{d^3 k}{(2\pi)^3} e^{ik \cdot x} \int_{-\infty}^{\infty} \frac{dk_0}{2\pi} e^{-ik_0 x_0} \frac{1}{k_0^2 - k^2} = -\tilde{D}_0(x)$$

use Residues to solve



2 integrals needed

$$\int_{-\infty}^{\infty} \frac{dk_0}{2\pi} e^{-ik_0 x_0} \frac{1}{k_0 \pm i\epsilon}$$

$$e^{-ik_0 x_0} = e^{-i(\text{Re } k_0 + i \text{Im } k_0)x_0} = e^{-i \text{Re } k_0 x_0} e^{-\text{Im } k_0 x_0}$$

distinguish cases: 1) $x_0 > 0 \rightarrow$ use $\text{Im } k_0 < 0$ i.e. γ_1

$$\Rightarrow e^{\text{Im } k_0 x_0} \rightarrow 0$$

2) $x_0 < 0 \rightarrow$ use $\text{Im } k_0 > 0$ i.e. γ_2

$$\int_{-\infty}^{\infty} \frac{dk_0}{2\pi} \rightarrow \lim_{R \rightarrow \infty} \int_{-R}^R \frac{dk_0}{2\pi} \text{ on the semi-circle } k_0 = R e^{i\varphi} \Rightarrow \text{Im } k_0 = R \sin \varphi$$

$\rightarrow$ exponential expression in R for $\varphi \neq 0, \pi$

$\Rightarrow$ semi circles will not contribute to the integral in the limit $R \rightarrow \infty$

$$I_{\pm}(x_0) = \int \frac{dk_0}{2\pi} e^{-ik_0 x_0} \frac{1}{k_0 \pm i\epsilon} \rightarrow \text{poles shifted to } k_0 = \pm i\epsilon \text{ i.e.}$$

Greens fct of $\square$ Electrodynamics

05/02

$$I_{\pm}(x_0) = \begin{cases} \oint_{\gamma_2} \frac{dk_0}{2\pi} e^{-ik_0 x_0} \frac{1}{k_0 \pm |\vec{k}| + i\epsilon}, & x_0 > 0 \\ \oint_{\gamma_1} \frac{dk_0}{2\pi} e^{-ik_0 x_0} \frac{1}{k_0 \pm |\vec{k}| - i\epsilon}, & x_0 < 0 \end{cases}$$

$x_0 > 0$ residue thm $\oint(\dots) = 0$ (no poles enclosed)

$x_0 < 0$ $\oint e^{-ik_0 x_0} \frac{1}{k_0 \pm |\vec{k}| - i\epsilon} = \frac{2\pi i}{2\pi} e^{-ix_0(\pm|\vec{k}| + i\epsilon)} = ie^{\pm i x_0 |\vec{k}|}$

combine cases using heaviside step function

$$\Theta(x_0) = \begin{cases} 1 & \text{if } x_0 > 0 \\ 0 & \text{otherwise} \end{cases}$$

$$I_{\pm}(x_0) = \Theta(-x_0) i e^{\pm i x_0 |\vec{k}|}$$

Recap: ~~$\square \delta_0(x) = \delta^{(4)}(x)$~~ $\square \delta_0(x) = \delta^{(4)}(x) \Rightarrow \square \Psi(x) = 4\pi f(x)$

$\Rightarrow \Psi(x) = \int d^4 x' \delta_0(x-x') 4\pi f(x')$

in fourier space: $\square e^{-ikx} = -k^2 e^{-ikx}$ $\chi(x), (\square \chi) = 0$

$\rightarrow -k^2 \tilde{D}_0(k) = 1$

$$D(x) = - \int \frac{d^3 \vec{k}}{(2\pi)^3} e^{i\vec{k} \cdot \vec{x}} \int_{-\infty}^{\infty} \frac{dk_0}{2\pi} e^{-ik_0 x_0} \frac{1}{k_0 \pm |\vec{k}|}$$

for downward shift of poles:

$$J_{\pm}(x_0) = I_{\pm}(x_0) \Big|_{\epsilon \rightarrow -\epsilon}$$

$$\Rightarrow J_{\pm}(x_0) = \Theta(\pm x_0) i e^{\pm i x_0 |\vec{k}|}$$

Electrodynamics

05/02

Define 2 Green's fnc's.

$$1) \underset{\text{advanced}}{D_{adv}}(x) = \int \frac{d^3k}{(2\pi)^3} e^{i\vec{k}\cdot\vec{x}} \frac{I_+(x_0) - I_-(x_0)}{2|\vec{k}|} = -\Theta(-x_0) \int \frac{d^3k}{(2\pi)^3} e^{i\vec{k}\cdot\vec{x}} \frac{\sin(|\vec{k}|x_0)}{|\vec{k}|}$$

$$2) \underset{\text{retarded}}{D_{ret}}(x) = \int \frac{d^3k}{(2\pi)^3} e^{i\vec{k}\cdot\vec{x}} \frac{I_+(x_0) - I_-(x_0)}{2|\vec{k}|} = \Theta(x_0) \int \frac{d^3k}{(2\pi)^3} e^{i\vec{k}\cdot\vec{x}} \frac{\sin(|\vec{k}|x_0)}{|\vec{k}|}$$

difference $D(x) = D_{ret}(x) - D_{adv}(x) = \left(\Theta(x_0) + \Theta(-x_0) \right) \int \frac{d^3k}{(2\pi)^3} e^{i\vec{k}\cdot\vec{x}} \frac{\sin(|\vec{k}|x_0)}{|\vec{k}|}$

$$\square D(x) = \square D_{ret}(x) - \square D_{adv}(x)$$

$$= \delta^{(4)}(x) - \delta^{(4)}(x) = 0$$

inverse spatial F.T

$$D_{ret}(x) = \frac{\Theta(x_0)}{(2\pi)^3} \int d^3k e^{i\vec{k}\cdot\vec{x}} \frac{\sin(|\vec{k}|x_0)}{|\vec{k}|}, \quad \vec{x} = |\vec{x}| \hat{x}, \quad \vec{k} \cdot \vec{x} = |\vec{k}| |\vec{x}| \cos\theta$$

$$= \frac{\Theta(x_0)}{(2\pi)^3} \int_0^\infty d|\vec{x}| |\vec{x}|^2 \int_{-1}^1 d(\cos\theta) \frac{1}{|\vec{x}|} e^{i|\vec{x}||\vec{k}|\cos\theta} \frac{1}{2i} (e^{i|\vec{k}|x_0} - e^{-i|\vec{k}|x_0})$$

$$\int_{-1}^1 d\cos\theta e^{i|\vec{x}||\vec{k}|\cos\theta} = \frac{1}{i|\vec{x}||\vec{k}|} (e^{i|\vec{x}||\vec{k}|} - e^{-i|\vec{x}||\vec{k}|})$$

$$\rightarrow D_{ret}(x) = \frac{\Theta_H}{(2\pi)^2} \frac{1}{|\vec{x}|} \int_0^\infty d|\vec{k}| (e^{i|\vec{x}||\vec{k}|} - e^{-i|\vec{x}||\vec{k}|}) (e^{i|\vec{k}|x_0} - e^{-i|\vec{k}|x_0})$$

$$= \frac{\Theta_H(x_0)}{8\pi^2 |\vec{x}|} \int_0^\infty d|\vec{k}| \left[e^{i|\vec{k}|(x_0+|\vec{x}|)} - e^{i|\vec{k}|(x_0-|\vec{x}|)} - e^{-i|\vec{k}|(x_0+|\vec{x}|)} + e^{-i|\vec{k}|(x_0-|\vec{x}|)} \right]$$

$$= -\frac{\Theta_H(x_0)}{8\pi^2 |\vec{x}|} \int_0^\infty d|\vec{k}| e^{i|\vec{k}|(x_0+|\vec{x}|)} - e^{i|\vec{k}|(x_0-|\vec{x}|)}$$

$$D_{ret} = -\frac{\Theta_H(x_0)}{8\pi^2 |\vec{x}|} \cdot 2\pi [\delta(x_0+|\vec{x}|) - \delta(x_0-|\vec{x}|)] = \frac{\Theta_H(x_0)}{4\pi |\vec{x}|} \delta(x_0-|\vec{x}|)$$

$$D_{adv} = -\frac{\Theta_H(-x_0)}{8\pi^2 |\vec{x}|} \cdot 2\pi [\delta(x_0+|\vec{x}|) - \delta(x_0-|\vec{x}|)] = \frac{\Theta_H(-x_0)}{4\pi |\vec{x}|} \delta(x_0+|\vec{x}|)$$

Electrodynamics 06/02

Green's function for $\square$

$$D_{\text{ret}}(x) = \frac{\Theta_4(x_0)}{4\pi|\vec{x}|} \delta(x_0 - |\vec{x}|) \quad , \quad D_{\text{adv}} = \frac{\Theta_4(x_0)}{4\pi|\vec{x}|} \delta(x_0 + |\vec{x}|)$$

used $\int_{-\infty}^{\infty} \frac{dx}{2\pi} e^{ix(x-a)} = \delta(x-a)$ use test function $l(x)$

$$\int_{-\infty}^{\infty} dx l(x) \int_{-\infty}^{\infty} \frac{dx}{2\pi} e^{-ix(x-a)} = \int \frac{dx}{2\pi} e^{ixa} \underbrace{\int_{-\infty}^{\infty} dx l(x) e^{-ixx}}_{\tilde{l}(x)}$$

$$= \int \frac{dx}{2\pi} e^{ixa} \tilde{l}(x) = l(a)$$

use $(x-x')$ as argument $\rightarrow D_{\text{ret}}(x-x') = \frac{\Theta_4(x_0-x'_0)}{4\pi|\vec{x}-\vec{x}'|} \delta(x_0-x'_0 - |\vec{x}-\vec{x}'|)$

$$\square \Psi = 4\pi f(x) \quad \Psi(x) = 4\pi \int D_{\text{ret}}(x-x') f(x') d^4x'$$

$$\square D_{\text{ret}}(x-x') = \delta^{(4)}(x-x')$$

Interpretation: disturbance at $(x'_0, \vec{x}')$ is felt at $(x_0, \vec{x})$

where $x_0 - x'_0 = |\vec{x} - \vec{x}'|$

$$t - t' = \frac{1}{c} |\vec{x} - \vec{x}'| \rightarrow \underbrace{t - \frac{1}{c} |\vec{x} - \vec{x}'|}_{\equiv t_{\text{ret}}} - t' = 0$$

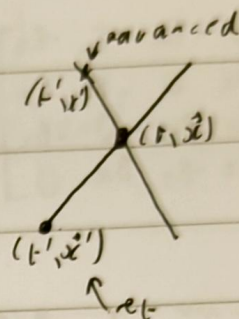
$$D_{\text{ret}}(\vec{x}-\vec{x}', t-t') = \frac{1}{c} \frac{\delta(t_{\text{ret}}-t')}{4\pi|\vec{x}-\vec{x}'|}$$

t_{ret} is the time it takes for a signal to travel the distance $|\vec{x}-\vec{x}'|$ at c
 t_{ret} is the time in the observer's frame at which the signal observed at t was created

Electrodynamics

Greens Fcn for $\square$

$$D_{adv}(\vec{x}-\vec{x}', t-t') = \frac{1}{c} \frac{\delta(t_{adv}-t')}{4\pi|\vec{x}-\vec{x}'|} \quad \text{where } t_{adv} \equiv t + \frac{1}{c}|\vec{x}-\vec{x}'|$$



Lorentz Covariance

$$\square F_{\mu\nu}(x) = \frac{4\pi}{c} (\partial_\mu j_\nu(x) - \partial_\nu j_\mu(x))$$

$$\text{Lorentz transf: } \tilde{x} = \lambda x$$

$$\tilde{F}_{\mu\nu}(\tilde{x}) = \lambda_\mu^\rho \lambda_\nu^\sigma F_{\rho\sigma}(x)$$

$$\tilde{j}_\mu(\tilde{x}) = \lambda_\mu^\rho j_\rho(x) \quad \left\{ \rightarrow \square \tilde{F}_{\mu\nu}(\tilde{x}) = \frac{4\pi}{c} (\tilde{\partial}_\mu \tilde{j}_\nu(\tilde{x}) - \tilde{\partial}_\nu \tilde{j}_\mu(\tilde{x})) \right.$$

$$\square = \partial_\mu \partial^\mu = \partial_\mu \partial^\mu \neq \tilde{\partial}_\mu \tilde{\partial}^\mu$$

not obvious
↓

$$\square \lambda_\mu^\nu \square D_{\mu\nu}(x) = \delta^{(4)}(x) \quad \text{expect } \square \tilde{D}_{\mu\nu}(\tilde{x}) = \delta^{(4)}(\tilde{x})$$

$$\delta^{(4)}(\tilde{x}) = \delta^{(4)}(\lambda x) = \frac{1}{\det \lambda} \delta^{(4)}(x)$$

Lorentz Transf of Electric Field

$$\text{L.T from } S \rightarrow S', \quad x'^\mu = \lambda^\mu_\nu x^\nu \quad \text{or} \quad x'_\mu = \lambda_\mu^\nu x_\nu$$

$$\lambda = (\lambda^\mu_\nu) \neq (\lambda_\mu^\nu) = \tilde{\lambda}$$

$$\lambda^\mu_\nu = g^{\mu\rho} g_{\nu\sigma} \lambda_\sigma^\rho = g^{\mu\rho} \lambda_\rho^\sigma g_{\nu\sigma} = g \tilde{\lambda} g^T$$

$$\tilde{\lambda} = g \lambda g^T = \begin{pmatrix} 1 & -1 & -1 & -1 \\ -1 & 1 & 1 & 1 \\ -1 & 1 & 1 & 1 \\ -1 & 1 & 1 & 1 \end{pmatrix} (\lambda^\mu_\nu) \begin{pmatrix} 1 & -1 & -1 & -1 \\ -1 & 1 & 1 & 1 \\ -1 & 1 & 1 & 1 \\ -1 & 1 & 1 & 1 \end{pmatrix}$$

$$= \left(\begin{array}{c|c} \lambda^0_0 & -\lambda^0_1, -\lambda^0_2, -\lambda^0_3 \\ \hline -\lambda^1_0 & (\lambda^k_l) \\ -\lambda^2_0 & \\ -\lambda^3_0 & \end{array} \right), \quad k, l = 1, 2, 3$$

Lorentz trans of E Field

Electrodynamics

07/02

$$\rightarrow F'^{\mu\nu}(x') = \lambda^\mu_\rho \lambda^\nu_\sigma F^{\rho\sigma}(x) \quad F \equiv (F^{\mu\nu})$$

$$F = \begin{bmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & -B_z & B_y \\ E_y & B_z & 0 & -B_x \\ E_z & -B_y & B_x & 0 \end{bmatrix}, \quad F'(x') = \lambda F(x) \lambda^T \rightarrow \begin{cases} \vec{E}' = \vec{F}_\lambda(\vec{E}, \vec{B}) \\ \vec{B}' = \vec{G}_\lambda(\vec{E}, \vec{B}) \end{cases}$$

recall $F^{\mu\nu} = \frac{1}{2} \epsilon^{\mu\nu\rho\sigma} F_{\rho\sigma}$ and $F'^{\mu\nu}(x') = \lambda^\mu_\rho \lambda^\nu_\sigma F^{\rho\sigma}(x)$

$$\det(F^{\mu\nu}) = (\det F^{\mu\nu}) \Big|_{(\vec{E}, \vec{B}) \rightarrow (\vec{B}', -\vec{E})}$$

$$\left. \begin{aligned} \vec{B}' &= \vec{F}_\lambda(\vec{B}, -\vec{E}) \\ -\vec{E}' &= \vec{G}_\lambda(\vec{B}, -\vec{E}) \end{aligned} \right\} \Rightarrow \begin{cases} F_\lambda(\vec{E}, \vec{B}) = -G_\lambda(\vec{B}, -\vec{E}) \\ G_\lambda(\vec{E}, \vec{B}) = F_\lambda(\vec{B}, -\vec{E}) \end{cases}$$

We only need $\vec{F}_\lambda$ to know how $\vec{E}$ & $\vec{B}$ transform

Jackson ch 11

observe $(F^{\mu\nu}) \Big|_{(\vec{E}, \vec{B}) \rightarrow (\lambda \vec{E}, \lambda \vec{B})} = \lambda (F^{\mu\nu}), \lambda \neq 0$

$$\rightarrow \vec{F}_\lambda(\lambda \vec{E}, \lambda \vec{B}) = \lambda \vec{F}_\lambda(\vec{E}, \vec{B})$$

consider a Lorentz boost in direction $\frac{\vec{p}}{p}$

$$\lambda = e^{-\vec{\zeta} \cdot \vec{k}}, \quad K_{1,2,3} \text{ are Lorentz boost generators}$$

$$K_1 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad K_2 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad K_3 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \quad \vec{\zeta} \parallel \vec{p}, \quad p = \tanh \zeta$$

$$\lambda^0_0 = \gamma$$

$$\lambda^k_k = 1 + \frac{\gamma-1}{\beta^2} \beta_k^2 \quad k=1,2,3$$

$$\lambda^0_k = -\gamma \beta_k = \lambda^k_0$$

$$\lambda^k_l = \frac{\gamma-1}{\beta^2} \beta_k \beta_l \quad k \neq l$$

$$\begin{aligned} \gamma^2 &= (1-\beta^2)^{-1} = \frac{1}{1-\tanh^2 \zeta} \\ &= \frac{1}{1-\frac{\sinh^2 \zeta}{\cosh^2 \zeta}} = \frac{\cosh^2 \zeta}{\cosh^2 \zeta - \sinh^2 \zeta} \\ &= \cosh^2 \zeta \end{aligned}$$

$$\gamma = \cosh \zeta$$

$$\beta \gamma = \sinh \zeta$$

Electrodynamics

determine $\vec{F}_A(\vec{E}, \vec{B})$

$$\vec{E}' = \gamma(\vec{E} + \vec{\beta} \times \vec{B}) - \frac{\gamma^2}{\gamma+1} \vec{\beta}(\vec{\beta} \cdot \vec{E}) = \vec{F}_A(\vec{E}, \vec{B})$$

$$\vec{B}' = \gamma(\vec{B} - \vec{\beta} \times \vec{E}) - \frac{\gamma^2}{\gamma+1} \vec{\beta}(\vec{\beta} \cdot \vec{B}) = -\vec{F}_A(\vec{B}, -\vec{E})$$

These can easily be inverted $\vec{\beta} \rightarrow -\vec{\beta}$ exchanges $S' \leftrightarrow S$

$$\vec{E} = \gamma(\vec{E}' - \vec{\beta} \times \vec{B}') - \frac{\gamma^2}{\gamma+1} \vec{\beta}(\vec{\beta} \cdot \vec{E}')$$

Assume a field of a charge at rest in $S' \rightarrow \vec{B}' = 0$

$$\Rightarrow \vec{E} = \gamma \vec{E}' - \frac{\gamma^2}{\gamma+1} \vec{\beta}(\vec{\beta} \cdot \vec{E}')$$

$$\vec{B} = \gamma(\vec{\beta} \times \vec{E}') = \gamma(\vec{\beta} \times (\vec{E} + \frac{\gamma^2}{\gamma+1} \vec{\beta}(\vec{\beta} \cdot \vec{E})))$$

$$\rightarrow \vec{B} = \vec{\beta} \times \vec{E}$$

Lorentz invariants

- 1) $F_{\mu\nu}'(x') F'^{\mu\nu}(x') = F_{\mu\nu}(x) F^{\mu\nu}(x)$
 - 2) $F_{\mu\nu}'(x') F'^{\mu\rho}(x') = F_{\mu\nu}(x) F^{\mu\rho}(x)$
 - 3) $F_{\mu\nu}'(x') F'^{\mu\nu}(x') = F_{\mu\nu}(x) F^{\mu\nu}(x)$
- Lorentz scalar Fields

in terms of $\vec{E}$ & $\vec{B}$ field,

$$F_{\mu\nu} = g_{\mu\rho} g_{\nu\sigma} F^{\rho\sigma} \Rightarrow (F_{\mu\nu}) = g F g^T = F|_{\vec{E} \rightarrow -\vec{E}}$$

$$F_{\mu\nu} F^{\mu\nu} = \text{tr}(g F g F^T) \quad F^T = -F$$

$$= -\text{tr}(g F g \bar{F}) = \dots = -2(\vec{E}^2 - \vec{B}^2)$$

$$2) \rightarrow -F_{\mu\nu} F^{\mu\nu} = 2(\vec{E}^2 - \vec{B}^2)$$

$$3) \rightarrow F_{\mu\nu} F^{\mu\nu} = -4 \vec{E} \cdot \vec{B}$$

12/02

Electrodynamics

radiation by
General sourcesRetarded Green fnctn for d'Alembert op $\square$

$$\square D_{\text{ret}}(x) = \delta^{(4)}(x), D_{\text{ret}}(x) = \frac{1}{2\pi} \Theta_H(x_0) \delta(x^2) = \frac{1}{4\pi|\vec{x}|} \delta(x_0 - |\vec{x}|) \\ = \frac{1}{4\pi|\vec{x}|} \frac{1}{c} \delta\left(t - \frac{|\vec{x}|}{c}\right)$$

Use $D_{\text{ret}}(x)$ to obtain a solutionfor $A_{\text{ret}}^\mu(x)$ in Lorentz gauge

$$A_{\text{ret}}^\mu = \frac{4\pi}{c} \int d^4x' D_{\text{ret}}(x-x') j^\mu(x')$$

$$A_{\text{ret}}^\mu = (\phi, \vec{A}) = \frac{4\pi}{c} \int d^4x' \frac{1}{4\pi|\vec{x}-\vec{x}'|} \delta(x_0 - x'_0 - |\vec{x}-\vec{x}'|) j^\mu(x'_0, \vec{x}')$$

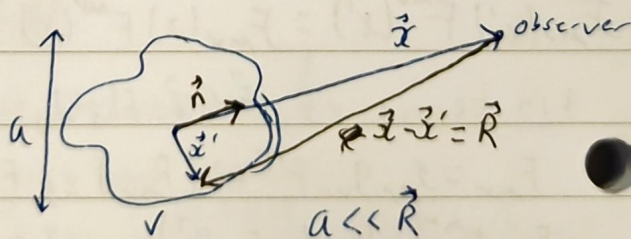
$$\vec{j} = (c\rho, \vec{j}) = \frac{4\pi}{c} \int d^3\vec{x}' \frac{1}{4\pi|\vec{x}-\vec{x}'|} j^\mu(x_0 - |\vec{x}-\vec{x}'|, \vec{x}') = \frac{1}{c} \int d^3\vec{x}' \frac{j^\mu(x_0 - |\vec{x}-\vec{x}'|, \vec{x}')}{|\vec{x}-\vec{x}'|}$$

$$\phi_{\text{ret}}(t, \vec{x}) = \int d^3\vec{x}' \frac{1}{|\vec{x}-\vec{x}'|} \rho\left(t - \frac{|\vec{x}-\vec{x}'|}{c}, \vec{x}'\right)$$

$$\vec{A}_{\text{ret}}(t, \vec{x}) = \frac{1}{c} \int d^3\vec{x}' \frac{\vec{j}}{|\vec{x}-\vec{x}'|} \left(\underbrace{t - \frac{|\vec{x}-\vec{x}'|}{c}}_{= t_{\text{ret}}}, \vec{x}' \right)$$

Assume that $\rho, \vec{j}$ are non-zero only inside a volume of linear extent a

$$\rightarrow \int d^3\vec{x}' \rightarrow \int_V d^3\vec{x}$$



$$\vec{x} = \vec{n} r, |\vec{x}| = r$$

$$|\vec{R}| = |\vec{x} - \vec{x}'| = |\vec{n} r - \vec{x}'| = r |\vec{n} - \frac{\vec{x}'}{r}|$$

$$= r \sqrt{(\vec{n} - \frac{\vec{x}'}{r})^2} = r \sqrt{1 - 2 \frac{\vec{n} \cdot \vec{x}'}{r} + \frac{x'^2}{r^2}}$$

$$\approx r \left(1 - \frac{\vec{n} \cdot \vec{x}'}{r} + O\left(\frac{a^2}{r^2}\right) \right)$$

$$= r - \vec{x}' \cdot \vec{n} + O\left(\frac{a^2}{r}\right)$$

13/02

Electrodynamics

radiation by
General sources

$$\frac{1}{|\vec{x} - \vec{x}'|} = \frac{1}{r} \left(1 - 2 \frac{\vec{x}' \cdot \vec{n}}{r} + \frac{\vec{x}'^2}{r^2} \right)^{-1/2} \approx \frac{1}{r} + \frac{\vec{x}' \cdot \vec{n}}{r^2}$$

to 1st order in $\frac{a}{r}$ where $\vec{A}_{\text{ret}}(t, \vec{x}) \approx \frac{1}{rc} \left(d\vec{x}' \cdot \vec{j}(t - \frac{r}{c} + \frac{\vec{n} \cdot \vec{x}'}{c}, \vec{x}') + \dots \right)$

Q: When can we Taylor expand $\vec{j}$ in its time argument

By assumption $\frac{\vec{x}' \cdot \vec{n}}{c} < \frac{a}{c}$, need an assumption on $\frac{\partial \rho}{\partial t}, \frac{\partial \vec{j}}{\partial t}$

$\rho(t, \vec{x}) = \int e^{i\omega t} \rho_{\omega}(\vec{x}) d\omega$ and call maxwell freq ω_{max} , ie $\rho_{\omega}(\vec{x}) \neq 0$ for $\omega > \omega_{\text{max}}$

require $\frac{a}{c} \left| \frac{1}{\rho} \frac{\partial \rho}{\partial t} \right| < 1 \Rightarrow \frac{a}{c} |\omega_{\text{max}}| < 1 \Leftrightarrow a \frac{2\pi}{\lambda_{\text{min}}} < 1$ or $\boxed{\lambda_{\text{min}} > a}$

in this case $\xrightarrow{\text{Taylor expand}} \rho(t_{\text{ret}}, \vec{x}') = \rho(t - \frac{r}{c}, \vec{x}') + \frac{\vec{n} \cdot \vec{x}'}{c} \dot{\rho}(t - \frac{r}{c}, \vec{x}') + \dots$ and analogous for $\vec{j}$

manipulate integral using $\dot{\rho} + \vec{\nabla} \cdot \vec{j} = 0$ $\leftarrow$ continuity eq

$$\partial_{\kappa} (j_{\kappa} x_{\kappa}) = (\partial_{\kappa} j_{\kappa}) x_{\kappa} + j_{\kappa} \underbrace{\partial_{\kappa} x_{\kappa}}_{=\delta_{\kappa\kappa}} = (\partial_{\kappa} j_{\kappa}) x_{\kappa} + j_{\kappa} \quad \text{summation over } \kappa \text{ implied}$$

$$\vec{j} = \partial_{\kappa} (j_{\kappa} \vec{x}) - (\vec{\nabla} \cdot \vec{j}) \vec{x}$$

$$\text{use w. t.h } \vec{x}' \rightarrow \vec{j}(t, \vec{x}') = -(\vec{\nabla} \cdot \vec{j}(t, \vec{x}')) \vec{x}' + \partial'_{\kappa} (j_{\kappa}(t, \vec{x}') \vec{x}')$$

$$\vec{A}_{\text{ret}}(t, \vec{x}) \approx \frac{1}{rc} \int d^3x' \left[\underbrace{-\vec{\nabla} \cdot \vec{j}(t - \frac{r}{c}, \vec{x}') \vec{x}'}_{=\vec{A}(t - \frac{r}{c}, \vec{x}')} + \underbrace{\partial'_{\kappa} (j_{\kappa}(t - \frac{r}{c}, \vec{x}') \vec{x}')}_{\text{Total derivative} \rightarrow 0} \right] + \dots$$

$$= \frac{1}{rc} \int d^3x' \dot{\rho}(t - \frac{r}{c}, \vec{x}') \vec{x}' + \dots$$

$$= \frac{1}{rc} \frac{d}{dt} \left(\underbrace{d^3x' \rho(t - \frac{r}{c}, \vec{x}') \cdot \vec{x}'}_{=\vec{d}(t - \frac{r}{c})} \right) = \frac{1}{rc} \dot{\vec{d}}(t - \frac{r}{c})$$

electric $\rightarrow$
dipole moment of
the charge density ρ

13/02

Electrodynamics

radiation by
accelerated fieldObtain $\vec{B}$ in this approximation (electric dipole approximation)

$$\vec{B} = \vec{\nabla} \times \vec{A} \approx -\frac{1}{rc} \vec{n} \times \dot{\vec{d}}(t-\frac{r}{c}) + \frac{1}{rc} \vec{n} \times \ddot{\vec{d}}(t-\frac{r}{c})(-\frac{1}{c})$$

Exercise: show that rotation sorts the components ~~as~~ given here
2 terms, which one dominates 1st term is down by $\frac{1}{r} \frac{c}{\omega_{\max}} = \frac{\lambda_{\max}}{2\pi r}$
2nd term dominates for $r \gg \lambda$, far field zone

Far field zone: $\vec{B} = -\frac{1}{rc} \vec{n} \times \ddot{\vec{d}}(t-\frac{r}{c})$

$$\vec{d}(t) = \int d^3x' \rho(t, \vec{x}')$$

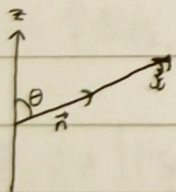
$$\frac{1}{c} \dot{\vec{E}} = \vec{\nabla} \times \vec{B} = \frac{1}{rc^2} \vec{n} \times (\vec{n} \times \ddot{\vec{d}}(t-\frac{r}{c})) + \frac{1}{rc^2} \vec{n} \times (\ddot{\vec{d}}(t-\frac{r}{c}))$$

$$\Rightarrow \vec{E} = \frac{1}{rc} \vec{n} \times (\vec{n} \times \ddot{\vec{d}}(t-\frac{r}{c})) = -\ddot{\vec{d}} \times \vec{n}$$

Radiated Power

Poynting Vector: $\vec{S} = \frac{c}{4\pi} \vec{E} \times \vec{B} = \frac{1}{4\pi c^2} |\vec{n} \times \ddot{\vec{d}}(t-\frac{r}{c})|^2 \vec{n}$, $\vec{S} \parallel \vec{n} = \frac{S}{r}$

Assume the dipole moment oscillates in z-direction $\rightarrow \vec{S} = \frac{|\ddot{d}|^2 \sin^2 \theta}{4\pi c^2} \vec{n}$



power maximised for $\theta = \frac{\pi}{2}, \frac{3\pi}{2}$
0 for $\theta = 0, \pi$

Total power Radiated

$$P = \int_{\text{surface element}} d^2r \cdot \vec{S}(r, \theta, \phi) = \frac{|\ddot{d}|^2}{4\pi c^2} \int_0^{2\pi} d\phi \int_0^\pi d\theta \sin^3 \theta \quad \left. \begin{array}{l} u = \cos \theta \\ du = -\sin \theta \end{array} \right\} \text{you should know this}$$

$$= \frac{8\pi}{3} \frac{|\ddot{d}|^2}{4\pi c^2} = \frac{2}{3c^2} |\ddot{d}|^2$$

$$d^2r = r^2 \sin \theta d\theta d\phi$$

non relativistic
Larmor formula

19/02

Electrodynamics

dipole
radiation by
fieldsrecap
reminder: Electric multipole moments of a charge distrib

$$\varphi(\vec{x}) = \int \frac{\rho(\vec{x}')}{|\vec{x} - \vec{x}'|} d^3x'$$

$$\frac{1}{|\vec{x} - \vec{x}'|} = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{2l+1} \frac{r_c^l}{r_s^l} Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi)$$

in our case: $|\vec{x}| \gg |\vec{x}'| \Rightarrow r_s = r, r_c = r'$ $\vec{x} \leftrightarrow (r, \phi, \theta)$, $\vec{x}' \leftrightarrow (r', \phi', \theta')$ spherical coords

Any function $g(\theta, \phi) = \sum_{l=0}^{\infty} \sum_{m=-l}^l A_{lm} Y_{lm}(\theta, \phi)$

ie g can be expanded in terms of spherical harmonics

$$A_{lm} = \int d\Omega Y_{lm}^*(\theta, \phi) g(\theta, \phi)$$

$$\bullet \text{Scalar product: } \int d\Omega Y_{lm}(\theta, \phi) Y_{l'm'}^*(\theta, \phi) = \delta_{ll'} \delta_{mm'}$$

$$\bullet \text{Completeness relation: } \sum_{l=0}^{\infty} \sum_{m=-l}^l Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi) = \delta(\phi - \phi') \delta(\cos \theta - \cos \theta')$$

$$\Rightarrow Y_{00} = \frac{1}{\sqrt{4\pi}}$$

$$Y_{11} = -\sqrt{\frac{3}{8\pi}} \sin \theta e^{i\phi}$$

Jackson ch 3

$$Y_{lm}^* = (-1)^m Y_{lm}(\theta, \phi)$$

$$\Phi(\vec{x}) = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{2l+1} \underbrace{\left[\int r'^l \rho(\vec{x}') Y_{lm}^*(\theta', \phi') d^3x' \right]}_{= q_{lm}: \text{multipole moments of } \rho} \frac{Y_{lm}(\theta, \phi)}{r^{l+1}}$$

$$\text{Cartesian coords: } q_{00} = \frac{1}{\sqrt{4\pi}} \int \rho(\vec{x}') d^3x' = \frac{Q}{\sqrt{4\pi}} \leftarrow \text{total charge}$$

$$q_{11} = \int \rho(\vec{x}') (x' - iy') d^3x' = -\sqrt{\frac{3}{8\pi}} (dx - idy)$$

$$q_{10} = \sqrt{\frac{3}{4\pi}} dz$$

$$q_{20} = \frac{1}{2} \sqrt{\frac{5}{4\pi}} Q_{33}$$

$$q_{21} = -\frac{1}{3} \sqrt{\frac{15}{8\pi}} (Q_{13} - iQ_{23})$$

$$q_{22} = \frac{1}{4} \sqrt{\frac{15}{2\pi}} (x' - iy')^2 \rho(\vec{x}') d^3x' = \frac{1}{12} \sqrt{\frac{15}{2\pi}}$$

radiation
of fields

Electrodynamics

19/02

where $Q_{ij} = \int (\partial_i x'_j - \partial_j x'_i) \rho(x') d^3x'$ $i, j = 1, 2, 3$

Q_{ij} symmetric, traceless

$$\Rightarrow \phi(x) = \frac{Q_{tot}}{r} + \frac{\vec{x} \cdot \vec{x}}{r^3} + \frac{1}{2} \sum_{i,j} Q_{ij} \frac{x_i x_j}{r^5}$$

back to radiation: $\vec{A}_{ret}(t, \vec{x}) = \frac{1}{c} \int d^3x' \frac{1}{|\vec{x} - \vec{x}'|} \vec{j}(t - \frac{|\vec{x} - \vec{x}'|}{c}, \vec{x}')$

$$= \frac{1}{rc} \int d^3x' \left(1 + \frac{\vec{x} \cdot \vec{x}'}{r^2} + \dots \right) \left[\vec{j}(t - \frac{r}{c}, \vec{x}') + \underbrace{\frac{\partial}{\partial t} \left(\vec{x} - \frac{r}{c} \vec{x}' \right) \frac{\vec{x} \cdot \vec{x}'}{rc}}_{O(\frac{a}{\lambda_{min}})} \right]$$

$O(\frac{a}{r})$ (neglect)

keep $\frac{a}{\lambda_{min}}$, neglect $O(\frac{a}{r})$

$$\rightarrow \vec{A}_{ret} = \vec{A}_{ret}(k, x) \Big|_{ED} + \frac{1}{r^2 c^2} \int d^3x' \vec{j}(t - \frac{r}{c}, \vec{x}') (\vec{x} \cdot \vec{x}')$$

Electric dipole

rewrite to get to the proper interpretation

$$\sum_{i,j} \partial_i (\partial_j x_i x_k) = \sum_i \left[(\partial_i j_i) x_i x_k + \overset{\delta_{ji}}{j_i} (\partial_i x_i) x_k + j_i x_i \overset{\delta_{ik}}{\partial_i} x_k \right]$$

$$= (\vec{\nabla} \cdot \vec{j}) x_i x_k + j_i x_k + j_k x_i$$

$$\sum_i \partial_i (\partial_i x'_i x'_k) = -\ddot{\rho} x'_i x'_k + \ddot{j}_i x'_k + \ddot{j}_k x'_i \quad \text{Contract with } \sum_{k=1}^3 n_k, \vec{n} = \frac{\vec{x}}{r}$$

$$\sum_i \partial_i (\partial_i x'_i (\vec{n} \cdot \vec{x}')) = -\ddot{\rho} x'_i (\vec{x}' \cdot \vec{n}) + \ddot{j}_i (\vec{x}' \cdot \vec{n}) + \vec{x}' (\vec{n} \cdot \ddot{\vec{j}}) = T_i$$

$$\Rightarrow \vec{T} = -\ddot{\rho} (\vec{x}' \cdot \vec{n}) \vec{x}' + \ddot{\vec{j}} \cdot (\vec{x}' \cdot \vec{n}) + \vec{x}' (\vec{n} \cdot \ddot{\vec{j}})$$

$$\textcircled{1} \ddot{\vec{j}} (\vec{x}' \cdot \vec{n}) = \vec{T} + \ddot{\rho} (\vec{x}' \cdot \vec{n}) \vec{x}' - (\ddot{\vec{j}} \cdot \vec{n}) \vec{x}'$$

We can derive a second identity considering $(\vec{n} \times (\ddot{\vec{j}} \times \vec{x}')) = c_k$

Consider $c_k = (\vec{n} \times (\ddot{\vec{j}} \times \vec{x}'))_k = \sum_{l,m} \epsilon_{klm} n_l (\ddot{\vec{j}} \times \vec{x}')_m$

$$= \sum_{l,m} \epsilon_{klm} n_l \sum_i \epsilon_{mij} \ddot{j}_i x'_j$$

19/02

Electrodynamics

radiation of
fields

$$\text{contd} = \sum_{i,j,k} \epsilon_{kmj} \epsilon_{mij} n_i \dot{x}_j = C_k$$

$$\sum_m \epsilon_{mki} \epsilon_{mij} = \delta_{ki} \delta_{lj} - \delta_{kj} \delta_{li}$$

$$\text{Wanted } \vec{C} = \vec{A}(\vec{n} \cdot \vec{x}') - \vec{x}'(\vec{n} \cdot \vec{j})$$

$$(2) \quad \vec{j}(\vec{n} \cdot \vec{x}') = \vec{C} + \vec{x}'(\vec{n} \cdot \vec{j})$$

$$\text{combine} \rightarrow \vec{j}(\vec{n} \cdot \vec{x}') = \frac{1}{2}(\textcircled{1} + \textcircled{2}) = \frac{1}{2} \vec{T} + \frac{1}{2} \vec{j}(\vec{x}' \cdot \vec{n}) - \frac{1}{2} \vec{x}'(\vec{n} \cdot \vec{j})$$

$$\frac{1}{2}(\textcircled{1} + \textcircled{2}) = \frac{1}{2} \vec{T} + \frac{1}{2} \vec{j}(\vec{x}' \cdot \vec{n}) - \frac{1}{2} \vec{x}'(\vec{n} \cdot \vec{j})$$

$$= \frac{1}{2} \vec{T} + \frac{1}{2} \vec{C} + \frac{1}{2} \vec{j}(\vec{x}' \cdot \vec{n}) \vec{x}'$$

total
derivative

$$\text{so } \int d^3x' \vec{j}(\vec{n} \cdot \vec{x}') = \int d^3x' \left[\frac{1}{2} \vec{j}(\vec{x}' \cdot \vec{n}) + \frac{1}{2} \int d^3x' \vec{n} \times (\vec{j} \times \vec{x}') \right]$$

$$\rightarrow \vec{A}_{\text{ret}}(t, \vec{x}) = \underbrace{\vec{A}_{\text{ret}}(t, \vec{x})}_{\text{electric dipole}} \Big|_{\text{ED}} + \underbrace{\vec{A}_{\text{ret}}(t, \vec{x})}_{\text{electric quadrupole}} \Big|_{\text{EQ}} + \underbrace{\vec{A}_{\text{ret}}(t, \vec{x})}_{\text{magnetic dipole}} \Big|_{\text{MD}} + \dots$$

Look at Magnetic Dipole

$$\vec{A}^{\text{MD}}(t, \vec{x}) = -\frac{1}{2rc^2} \vec{n} \times \int d^3x' \vec{x}' \times \vec{j}(t - \frac{r}{c}, \vec{x}')$$

$$\vec{A}^{\text{EQ}}(t, \vec{x}) = \frac{1}{2rc^2} \int d^3x' \ddot{\rho}(t - \frac{r}{c}, \vec{x}') (\vec{n} \cdot \vec{x}') d^3x'$$

magnetic dipole moment of a current density $\vec{j}$ in magnetostatics

$$\vec{m} = \frac{1}{2} \int d^3x' \vec{x}' \times \vec{j}(\vec{x}')$$

$$\rightarrow \vec{A}_{\text{ret}}^{\text{MD}}(t, \vec{x}) = -\frac{1}{rc^2} \vec{n} \times \vec{m}(t - \frac{r}{c})$$

radiation
of the field

Electrodynamics

19/02

Obtain corresponding $\vec{E}$ & $\vec{B}$ fields for MD

$$\vec{B}^{MD} = \vec{\nabla} \times \vec{A}^{MD} = \frac{1}{r^2 c^2} \vec{n} \times [\vec{n} \times \dot{\vec{m}}(t - \frac{r}{c})] + \underbrace{\frac{1}{r c^3} \vec{n} \times [\vec{n} \times \ddot{\vec{m}}(t - \frac{r}{c})]}_{\text{dominates in far field}}$$

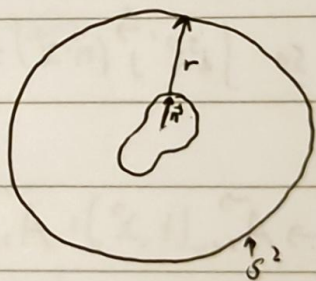
$$\vec{B}^{MD} \approx \frac{1}{r c^3} \vec{n} \times [\vec{n} \times \ddot{\vec{m}}(t - \frac{r}{c})], \text{ use } \vec{E}^{MD} = c \cdot \vec{\nabla} \times \vec{B}^{MD}$$

$$\vec{E}^{MD} = -\frac{1}{r c^3} \vec{n} \times (\underbrace{\vec{n} \times [\vec{n} \times \ddot{\vec{m}}(t - \frac{r}{c})]}_{= -\vec{n} \times \ddot{\vec{m}}}) = -\vec{n} \times \vec{B}^{MD} = \frac{1}{r c^3} \vec{n} \times \ddot{\vec{m}}(t - \frac{r}{c})$$

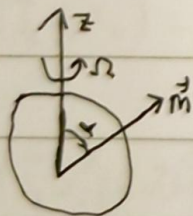
Power radiated

$$\vec{S}^{MD} = \frac{c}{4\pi} (\vec{E}^{MD} \times \vec{B}^{MD}) = \frac{c}{4\pi} |\vec{B}^{MD}|^2 \vec{n} = \frac{c}{4\pi} \frac{1}{r^2 c^6} |\ddot{\vec{m}}(t - \frac{r}{c})|^2 \sin^2 \theta \vec{n}$$

$$\text{total power: } P = \int_S d^2 \vec{r} \cdot \vec{S}^{MD} = \frac{|\ddot{\vec{m}}|^2}{4\pi c^5} 2\pi \frac{4}{3} = \frac{2}{3 c^5} |\ddot{\vec{m}}|^2$$



Pulsar Model



movement of magnetic moment

$$\vec{m} = \begin{pmatrix} \sin \alpha \cos(\Omega t) \\ \sin \alpha \sin(\Omega t) \\ \cos \alpha \end{pmatrix} \Rightarrow \ddot{\vec{m}} = -m \sin \alpha \Omega^2 \begin{pmatrix} \cos(\Omega t) \\ \sin(\Omega t) \\ 0 \end{pmatrix}$$

$$P = \frac{2}{3 c^5} |\ddot{\vec{m}}|^2 = \frac{2 m^2 \sin^2 \alpha \Omega^4}{3 c^5}$$

rotational energy of rotating star $E = \frac{1}{2} I \Omega^2$, $I = \frac{2}{5} M R^2$

$$\rightarrow P = -\dot{E} = -\frac{1}{2} I \Omega \dot{\Omega}$$

radiation of
the field

Electrodynamics

20/02

assume $\vec{B}$ -field at the surface of the star is given by $\vec{B}^{MO} \Rightarrow |\vec{B}^{MO}|$

typical value for field $\sim 10^8 \text{ T}$ (strongest field in lab $\sim 100 \text{ T}$)

$$\Omega = 2\pi \cdot 30/s$$

$$\dot{E} = -10^{32} \text{ J/s}$$

$$M = 14M_0 = 3 \times 10^{30} \text{ kg}$$

$$\Rightarrow \dot{\Omega} = -2 \times 10^{-9} / s^2$$

$$R = 15 \text{ km}$$

$\vec{B}$ -field of a magnetic dipole

$$\vec{B}(\vec{x}) = \frac{1}{cr^3} (3(\vec{m} \cdot \vec{n})\vec{n} - \vec{m}), \quad \vec{n} = \frac{\vec{x}}{r}, \quad r = |\vec{x}|$$

$$B_{\text{max}} \text{ for } \vec{R} \parallel \vec{B} \rightarrow |\vec{B}_{\text{max}}| = \frac{2m_0}{cr^3}$$

Electric Quadrupole

$$\vec{A}_{\text{ret}} \approx \vec{A}^{E0} + \vec{A}^{M0} + \vec{A}^{EQ}$$

$$Q_{ij} = \int d^3x' \rho(x') (3x'_i x'_j - r'^2 \delta_{ij}) \quad \left. \begin{array}{l} \text{Traceless} \Rightarrow \sum Q_{ii} = 0 \\ \text{Symmetric} \Rightarrow Q_{ij} = Q_{ji} \end{array} \right\}$$

$$\begin{aligned} \vec{A}_k^{EQ}(t, \vec{x}) &= \frac{1}{2rc^2} \sum_i \int d^3x' x'_k x'_i n_i \ddot{\rho}(t - \frac{r}{c}, \vec{x}') \\ &= \frac{1}{6rc^2} \sum_i n_i \underbrace{\left(\int d^3x' (3x'_k x'_i - r'^2 \delta_{ki}) \ddot{\rho}(t - \frac{r}{c}, \vec{x}') \right)}_{\rightarrow \ddot{Q}_{ki}} \end{aligned}$$

$$\begin{aligned} \vec{A}_k^{EQ} &= \frac{1}{6rc^2} \sum_i n_i \left(\ddot{Q}_{ki}(t - \frac{r}{c}) + \int d^3x' r'^2 \delta_{ki} \ddot{\rho}(t - \frac{r}{c}, \vec{x}') \right) \\ &= \frac{1}{6rc^2} \left[\sum_i n_i \ddot{Q}_{ki} + n_k \underbrace{\left(\int d^3x' r'^2 \ddot{\rho}(t - \frac{r}{c}, \vec{x}') \right)}_{\text{some function of } t, r} \right] \end{aligned}$$

$$\begin{aligned} (\nabla \times \vec{A})_k &= \sum_{i,j} \epsilon_{kij} \partial_i A_j \\ \partial_i \left(\frac{x_i}{r} \right) &= \frac{\delta_{ii}}{r} - \frac{x_i}{r^2} \partial_i r \\ &= \frac{\delta_{ii}}{r} - \frac{x_i x_i}{r^3} \\ &\quad \text{symmetric under } i \leftrightarrow j \end{aligned}$$

The additional term is radial $\rightarrow A \propto \vec{n} \propto \vec{x}$

The curl of a radial term vanishes $\xrightarrow{\text{Sho-n-ho-e}}$

$\Rightarrow$ when we contract with ϵ_{ijk} it will be 0

radiation of
Field

Electrodynamics

21/02

$$\vec{A}^{EQ}(t, \vec{x}) = \frac{1}{6rc^2} \vec{n} \cdot \vec{Q}(t - \frac{r}{c})$$

$\vec{E}^{EQ}, \vec{B}^{EQ}$ fields

$$\vec{B}^{EQ} = -\frac{1}{6rc^2} \vec{n} \times [\vec{n} \cdot \vec{Q}(t - \frac{r}{c})], \quad \vec{E}^{EQ} = -\vec{n} \times \vec{B}^{EQ}(t, \vec{x})$$

So $\vec{E} \propto \vec{E}^{EO} + \vec{E}^{MO} + \vec{E}^{EQ}$

$$\vec{B} \approx \vec{B}^{EO} + \vec{B}^{MO} + \vec{B}^{EQ}$$

$$\text{and } \vec{J} = \frac{c}{4\pi} (\vec{E} \times \vec{B}) \approx \frac{c}{4\pi} (\vec{E}^{EO} + \vec{E}^{MO} + \vec{E}^{EQ}) \times (\vec{B}^{EO} + \vec{B}^{MO} + \vec{B}^{EQ})$$

Contains mixed terms such as $\frac{c}{4\pi} (\vec{E}^{EO} \times \vec{B}^{MO})$ but such mixed terms vanish when considering total power radiated

Potential of Moving Point Charge

$$\square A^\mu = \frac{4\pi}{c} j^\mu, \text{ Lorentz Gauge } \partial_\mu A^\mu = 0$$

$$A_{ret}^\mu(x) = \frac{4\pi}{c} \int d^4x' D_{ret}(x-x') j^\mu(x')$$

what is j^μ for a point charge?

trajectory $t \rightarrow \vec{r}(t)$ velocity $t \rightarrow \dot{\vec{r}}(t) = \vec{v}(t)$

$$j^\mu(x) = (c\rho(t, \vec{x}), \vec{j}(t, \vec{x}))$$

$$\rho(t, \vec{x}) = q \delta^{(3)}(\vec{x} - \vec{r}(t))$$

$$\vec{j}(t, \vec{x}) = q \vec{v}(t) \delta^{(3)}(\vec{x} - \vec{r}(t))$$

$$\rightarrow j^\mu(x) = \underbrace{(c, \vec{v}(t))}_{= \frac{1}{\gamma} u^\mu} q \delta^{(3)}(\vec{x} - \vec{r}(t)), \quad \gamma^2 = (1 - \vec{\beta}^2)^{-1}, \quad \vec{\beta} = \frac{\vec{v}}{c}$$

introduce: $r^0(\tau) = c\tau$, $r^k(\tau) = \vec{r}_k(t)$, $k=1,2,3$

τ = proper time of point charge

Potential of moving point charge Electrodynamics 21/02

recall: $ds^2 = c^2 dt^2 - |d\vec{x}|^2 = c^2 dt^2 (1 - \vec{\beta}^2) = \frac{c^2 dt^2}{\gamma^2}$
 $= c^2 d\tau^2 \Rightarrow d\tau = \frac{dt}{\gamma}$

$$J^\mu(x) = q \int \frac{dt}{\gamma(t)} \delta(t - \frac{r_0(t)}{c}) \prod_{k=1}^3 \delta(\vec{x}_k - \vec{r}^k(t))$$

$$= q \cdot c \int d\tau \prod_{k=0}^3 \delta(x^\mu - r^\mu(\tau)) U^\mu(\tau)$$

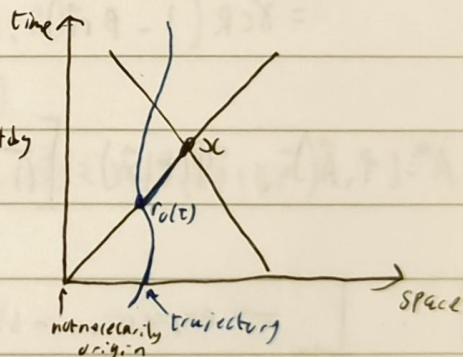
$\Rightarrow \text{Covariant } J^\mu = q c \int d\tau \delta^{(4)}(x - r(\tau)) U^\mu(\tau)$

$$A^\mu(x) = \frac{4\pi}{c} \int d^4x' D_{ret}(x - x') J^\mu(x') = 4\pi q_0 \left(d^4x' \frac{1}{2\pi} \Theta_H(x_0 - x'_0) \delta[(x - x')^2] \right. \\ \left. \times \int d\tau \delta^{(4)}(x - r(\tau)) U^\mu(\tau) \right)$$

$$= 2q \int d\tau \Theta_H(x_0 - r_0(\tau)) \delta[(x - r(\tau))^2] U^\mu(\tau)$$

So one only gets contributions to $A^\mu(x)$

only for $(x - r(\tau_0))^2 = 0 \rightarrow$ defines τ_0 implicitly



$\rightarrow$ carry out the τ integral

$$\delta(f(x)) = \sum \frac{\delta(x - x_i)}{|f'(x_i)|}$$

$$\Theta_H(x_0 - r_0(\tau)) \delta[(x - r(\tau))^2] = \frac{\delta(\tau - \tau_0)}{| \frac{dx}{d\tau}(\tau_0) |}$$

$$\frac{df}{d\tau} = \frac{d}{d\tau} (x - r(\tau))^2 = 2(x - r(\tau)) \frac{d}{d\tau} (-r(\tau)) = -2U^\mu(\tau) (x - r_\mu(\tau))$$

$$\Rightarrow A^\mu(x) = \int d\tau \delta(\tau - \tau_0) \frac{x_\mu U^\mu(\tau)}{|x - r(\tau)|_\rho U^0(\tau)}$$

$\rightarrow A^\mu(x) = \frac{q U^\mu(\tau_0)}{(x - r(\tau_0))_\rho U^0} \quad \text{Lienard-Wiechert potential}$

26/02

Electrodynamics

Radiation by a
moving point
charge

$$A^\mu(x) = \frac{q U^\mu(\tau_0)}{U_r(\tau_0) [x - r(\tau_0)]^2} \quad \text{Lienard-Wiechert potential}$$

trajectory: $r^\mu(\tau)$, τ = proper time of particle $d\tau = \frac{dt}{\gamma(t)}$

$$r^0(\tau) = ct$$

relation $t \leftrightarrow \tau$

$$r^k(\tau) = \vec{r}_k(t), k=1,2,3$$

$$ds^2 = c^2 d\tau^2 - |dx|^2 = c^2 dt^2 (1 - \beta^2) \\ = \frac{c^2 dt^2}{\gamma^2} = c^2 d\tau^2$$

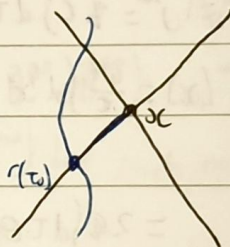
back to 3-vector

$$(x - r(\tau_0))^2 = 0 \Rightarrow x_0 - r(\tau) = |\vec{x} - \vec{r}(\tau)| \equiv R$$

$$U_r(x-r)^\mu = U_0(x_0 - r_0(\tau_0)) - \vec{U}(\vec{x} - \vec{r}(\tau_0))$$

$$= \gamma c R - \gamma \vec{v} \cdot \vec{n} R$$

$$= \gamma c R (1 - \beta \vec{n})$$



$$\rightarrow A^\mu = (\phi, \vec{A}), \quad \phi(t, \vec{x}) = \left[\frac{q}{(1 - \beta \cdot \vec{n}) R} \right]_{\text{ret}} \quad \vec{A}(t, \vec{x}) = \left[\frac{q \vec{\beta}}{(1 - \beta \cdot \vec{n}) R} \right]_{\text{ret}}$$

$\left[\dots \right]_{\text{ret}} \Rightarrow$ expression in brackets to be evaluated @ t_{ret}

$$r_0(\tau_0) = x_0 - R = c \cdot t_{\text{ret}}, \quad t_{\text{ret}} = t - \frac{R}{c}$$

Obtain $\vec{E}, \vec{B}$ fields by calculating $F_{\mu\nu}$ from A_μ $F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$

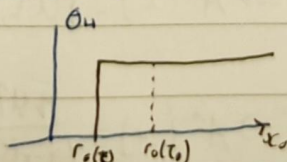
$$A_\mu = 2q \int d\tau U^\mu(\tau) \theta(x_0 - r^0(\tau)) \delta((x - r(\tau))^2)$$

$$\partial^\nu A^\mu = 2q \int d\tau U^\mu(\tau) \theta_\nu(x_0 - r_0(\tau)) \partial^\nu \delta((x - r(\tau))^2)$$

assume $x_0 - r_0(\tau) = R \neq 0$

$$\partial^\nu \delta(f) = \partial^\nu f \frac{d}{df} \delta(f) = \partial^\nu f \frac{d\tau}{df} \frac{d}{d\tau} \delta(f)$$

$$f = (x - r(\tau))^2$$



$$\rightarrow \begin{cases} \partial^\nu f = 2(x - r(\tau))^\nu \\ \frac{\partial f}{\partial \tau} = -2(x - r(\tau))^\mu \frac{dr(\tau)^\mu}{d\tau} \\ \quad \quad \quad = -2(x - r(\tau))^\mu U^\mu(\tau) \end{cases}$$

26/02

Electrodynamics

radiation by
a moving point
charge

$$\dots \Rightarrow \partial^\nu \delta(t) = \frac{-(x-r(t))^\nu}{(x-r(t))_\rho U^\rho(t)} \frac{d}{dt} \delta(t)$$

$$\Rightarrow \partial^\nu A^\mu = 2q \int dt U^\mu(t) \Theta_H(x_0 - r_0(t)) \left\{ -\frac{(x-r(t))^\nu}{(x-r(t))_\rho U^\rho(t)} \right\} \frac{d}{dt} \delta[(x-r(t))^2]$$

integrate by parts. Surface term vanishes due to δ function

$$\partial^\nu A^\mu = 2q \int dt \delta[(x-r(t))] \Theta_H(x_0 - r_0(t)) \frac{d}{dt} \left\{ \frac{(x-r(t))^\nu U^\mu(t)}{(x-r(t))_\rho U^\rho(t)} \right\} U^\mu(t)$$

$$F^{\mu\nu}(x) = \partial^\mu A^\nu - \partial^\nu A^\mu = 2q \int dt \delta[(x-r(t))] \Theta_H(x_0 - r_0(t)) \frac{d}{dt} \left\{ \frac{(x-r(t))^\mu U^\nu(t) - (x-r(t))^\nu U^\mu(t)}{(x-r(t))_\rho U^\rho(t)} \right\}$$

$$\text{Carry out } \tau \text{ integral} \Rightarrow \frac{q}{U_0(t)(x-r(t))^\sigma} \frac{d}{dt} \left\{ \frac{(x-r(t))^\mu - (x-r(t))^\nu U^\mu(t)}{(x-r(t))_\rho U^\rho(t)} \right\}_{\tau=\tau_0}$$

Specify indices $U \vec{F}^{k0} = \vec{E}_K$

$$(x-r(t))^\mu = (R, R\hat{n}) = R(1, \hat{n})$$

$$U^\mu = \gamma(c, \vec{v}) = \gamma c(1, \vec{\beta})$$

$$\frac{dU^\mu}{dt} = c \left(\frac{d\gamma}{dt}, \frac{d(\gamma\vec{\beta})}{dt} \right) = \gamma c (\dot{\gamma}, (\dot{\gamma}\vec{\beta}) + \gamma\dot{\vec{\beta}}) = \gamma c (\dot{\gamma}, \dot{\gamma}\vec{\beta} + \dot{\vec{\beta}})$$

$$\rightarrow \frac{d}{dt} U_\sigma (x-r(t))^\sigma = \frac{dU_\sigma}{dt} (x-r(t))^\sigma - U_\sigma \frac{d}{dt} (x-r(t))^\sigma =$$

$$\vec{E} \equiv F^{k0} = \left[\frac{q(R-\rho)}{R^3 \gamma^2 (1-\vec{\beta}\hat{n})^3} + \frac{q[(R-\vec{\rho})\dot{\vec{\beta}}\hat{n} - \vec{\rho}(\hat{n}\cdot\dot{\vec{\beta}})]}{Rc(1-\vec{\beta}\hat{n})^3} \right]_{\text{ret}}$$

side calculation: $(\vec{n} \times \{(\vec{n}-\vec{\rho}) \times \dot{\vec{\rho}}\}) = \epsilon_{klm} n_l ((\vec{n}-\vec{\rho}) \times \dot{\vec{\rho}})_m = \epsilon_{klm} \{ \epsilon_{mij} n_l ((\vec{n}-\vec{\rho})_i \dot{\rho}_j) \}$

$$\epsilon_{klm} \epsilon_{mij} = \epsilon_{mkl} \epsilon_{mij} = \delta_{ki} \delta_{lj} - \delta_{li} \delta_{kj}$$

$$= \delta(\vec{n}-\vec{\rho})_k \vec{n} \cdot \dot{\vec{\rho}} - \dot{\vec{\rho}}_k \vec{n} \cdot (\vec{n}-\vec{\rho})$$

27/02

Electrodynamics

rad. at. on by a moving point charge

$$\vec{E}(\vec{x}, t) = \underbrace{\left[\frac{q(\vec{n} - \vec{\beta})}{R^2 \gamma^3 (1 - \vec{\beta} \cdot \vec{n})^3} \right]_{\text{ret}}}_{\text{velocity field} \sim \frac{1}{R^2}} + \underbrace{\left[\frac{\frac{q}{c} \{ \vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}] \}}{R (1 - \vec{\beta} \cdot \vec{n})^3} \right]_{\text{ret}}}_{\text{acceleration field} \sim \frac{1}{R}}$$

$$\vec{B}(\vec{x}, t) = [\vec{n} \times \vec{E}]_{\text{ret}} \quad \text{recall Poynting vector, } \vec{E} \times \vec{B} \sim \frac{1}{R^2} \text{ for acceleration fields}$$

$\vec{B} = \vec{0} \Rightarrow$ only velocity fields $\Rightarrow$ must be transformation of $\vec{E}$ field of a point charge

$$\vec{S} = \frac{c}{4\pi} \vec{E}_{\text{rad}} \times \vec{B}_{\text{rad}} = \frac{c}{4\pi} [\vec{n} |\vec{E}_{\text{rad}}|^2]_{\text{ret}}$$

$$[\vec{S} \cdot \vec{n}]_{\text{ret}} = \text{radiated energy per time and area measured at time } t \text{ and radiated at } t_{\text{ret}} = t - \frac{R(t_{\text{ret}})}{c}$$

$$t = T_2 + \frac{R(T_2)}{c}$$

$$\int_{t=T_1 + \frac{R(T_1)}{c}}^{t=T_2 + \frac{R(T_2)}{c}} dt [\vec{S} \cdot \vec{n}]_{\text{ret}} = \text{energy radiated at retarded time interval } T_2 - T_1$$

$$= \int_{T_1}^{T_2} dt_{\text{ret}} \underbrace{\vec{S} \cdot \vec{n} \frac{dt}{dt_{\text{ret}}}}_{\text{relevant combination}}$$

Define: $\frac{dP(t_{\text{ret}})}{d\Omega} = R^2 (\vec{S} \cdot \vec{n}) \frac{dt}{dt_{\text{ret}}}$

$$= \frac{d}{dt} \left(t - \frac{R}{c} \right) = \frac{d}{dt} \left(t - \frac{1}{c} |\vec{x} - \vec{r}(t_{\text{ret}})| \right)$$

$$= \frac{1}{c} \frac{d}{dt} \left(|\vec{x} - \vec{r}(t_{\text{ret}})| \right) = 1 - \frac{1}{c} \frac{\vec{x} \cdot (\vec{x} - \vec{r}(t_{\text{ret}})) \cdot \frac{d}{dt} (-\vec{r}(t_{\text{ret}}))}{|\vec{x} - \vec{r}(t_{\text{ret}})|^2}$$

$$= 1 - \frac{1}{c} \frac{(\vec{x} - \vec{r}(t_{\text{ret}})) \cdot (-\dot{\vec{r}}(t_{\text{ret}})) \frac{dt_{\text{ret}}}{dt}}{|\vec{x} - \vec{r}(t_{\text{ret}})|^2}$$

$$= 1 - \frac{1}{c} (-\vec{n} \cdot \dot{\vec{r}}) \frac{dt_{\text{ret}}}{dt} = 1 + \vec{\beta} \cdot \vec{n} \frac{dt_{\text{ret}}}{dt}$$

$$\frac{dt_{\text{ret}}}{dt} (1 - \vec{\beta} \cdot \vec{n}) = 1 \Rightarrow \frac{dt}{dt_{\text{ret}}} = \boxed{1 - [\vec{\beta} \cdot \vec{n}]_{\text{ret}}}$$

27/02

Electrodynamics

radiation by a moving point charge

$$\vec{S} = \frac{c}{4\pi} \vec{E}_{rad} \times \vec{B}_{rad} = \frac{c}{4\pi} [\vec{n} |E_{rad}|^2]_{ret}$$

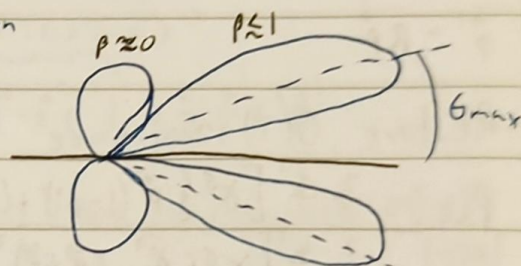
$$\frac{d\vec{P}(t_{ret})}{d\Omega} = \frac{q^2}{4\pi c} \frac{|\vec{n} \times (\dot{\vec{n}} - \dot{\vec{\beta}}) \times \dot{\vec{\beta}}|^2}{(1 - \dot{\vec{\beta}} \cdot \vec{n})^5}$$

Example rectilinear motion $\dot{\vec{\beta}} \parallel \vec{\beta} \Rightarrow \dot{\vec{\beta}} \times \vec{\beta} = 0$

$$\frac{d\vec{P}(t_{ret})}{d\Omega} = \frac{q^2 \dot{\vec{\beta}}^2 \sin^2 \theta}{4\pi c (1 - \beta \cos \theta)^5}$$

Choose $\vec{\beta}$ in z direction
 θ angle between $\vec{n}$ & $\vec{\beta}$ = $f(\theta)$ distribution
angular ~~density~~ of radiation $\beta \rightarrow 0, f(\theta) \sim \sin^2 \theta \rightarrow$ electric dipole

$$\beta \rightarrow 1, \frac{df(\theta)}{d\theta} = 0 \Rightarrow \theta_{max} \rightarrow \frac{1}{\gamma}$$



$$\vec{P}(t_{ret}) = \frac{q^2}{4\pi c} \int d\Omega \frac{|\vec{n} \times (\dot{\vec{n}} - \dot{\vec{\beta}}) \times \dot{\vec{\beta}}|^2}{(1 - \dot{\vec{\beta}} \cdot \vec{n})^5}$$

$$\text{simplify } (\vec{n} \times (\dot{\vec{n}} - \dot{\vec{\beta}}) \times \dot{\vec{\beta}})^2 = \dot{\beta}^2 (1 - \dot{\vec{\beta}} \cdot \vec{n})^2 + 2(\dot{\vec{\beta}} \cdot \vec{n})(\dot{\vec{\beta}} \cdot \dot{\vec{n}})(1 - \dot{\vec{\beta}} \cdot \vec{n}) - \dot{\vec{n}} \cdot \dot{\vec{\beta}}$$

$$\vec{P}(t_{ret}) = \frac{q^2}{4\pi c} \left[\int d\Omega \frac{\dot{\beta}^2}{(1 - \dot{\vec{\beta}} \cdot \vec{n})^2} + \int d\Omega \frac{2(\dot{\vec{\beta}} \cdot \vec{n})(\dot{\vec{\beta}} \cdot \dot{\vec{n}})}{(1 - \dot{\vec{\beta}} \cdot \vec{n})^4} - \gamma^{-2} \int d\Omega \frac{(\dot{\vec{n}} \cdot \dot{\vec{\beta}})^2}{(1 - \dot{\vec{\beta}} \cdot \vec{n})^5} \right]$$

$$\text{coordinates } \vec{\beta} = (0, 0, \beta), \vec{n} = (\cos \phi \sin \theta, \sin \phi \sin \theta, \cos \theta)$$

$$\Rightarrow \begin{cases} \vec{n} \cdot \vec{\beta} = \beta \cos \theta \\ \vec{n} \cdot \dot{\vec{\beta}} = \dot{\beta}_1 \cos \phi \sin \theta + \dot{\beta}_2 \sin \phi \sin \theta + \dot{\beta}_3 \cos \theta \end{cases}$$

$$d\Omega = \sin \theta d\theta d\phi$$

$$\phi \text{ integrals across: } \int_0^{2\pi} d\phi \vec{n} \cdot \dot{\vec{\beta}} = 2\pi \dot{\beta}_3 \cos \theta, \quad \int_0^{2\pi} d\phi (\vec{n} \cdot \dot{\vec{\beta}})^2 = \pi \dot{\beta}_1^2 (1 - \cos^2 \theta) + \pi \dot{\beta}_2^2 (\cos^2 \theta - 1)$$

$$I_n^m(\rho) = \int dx \frac{x^m}{(1 - \rho x)^n}$$

$$\rightarrow \vec{P}(t_{ret}) = \frac{q^2}{4\pi c} \left\{ 2\pi \dot{\beta}^2 I_3^0(\rho) + 4\pi (\dot{\vec{\beta}} \cdot \dot{\vec{\beta}}) \dot{\beta}_3 I_3^1(\rho) - \gamma^{-2} \pi \dot{\beta}^2 (I_5^0(\rho) - I_5^2(\rho)) - \gamma^{-2} \pi \dot{\beta}_1^2 (I_5^2(\rho) - I_5^0(\rho)) \right\}$$

28/02

Electrodynamics

radiation by a moving point charge

$$\text{Contd. } \left. \begin{aligned} I_3^0(\beta) &= 2\gamma^4 & I_3^6(\beta) &= 2\gamma^8(\beta^2+1) \\ I_4^1(\beta) &= \frac{8}{3}\beta\gamma^6 & I_5^2(\beta) &= \frac{2}{3}\gamma^8(1+5\beta^2) \end{aligned} \right\} \text{ by Wolfram}$$

3D rotation

$$R^T R = \mathbb{1}_{3 \times 3}$$

$$\vec{n}' = R \vec{n} \quad \rightarrow \quad \vec{n}' \cdot \vec{\beta}' = \vec{n}'^T \vec{\beta}' = \vec{n}^T R^T R \vec{\beta} = \vec{n}^T \vec{\beta} \rightarrow \text{invariant}$$

$$\vec{\beta}' = R \vec{\beta}$$

$$\vec{p}' = R \vec{p}$$

Restore $O(3)$ invariance $\beta \dot{\beta}_i \rightarrow \vec{\beta} \cdot \dot{\vec{\beta}}$

$$P(t) = \frac{2}{3} \frac{q^2}{c} \left[\gamma^6 \{ \dot{\vec{\beta}}^2 (1 - \beta^2) + (\vec{\beta} \cdot \dot{\vec{\beta}})^2 \} \right]_{\text{ret}}$$

$$P(t) = \frac{2}{3} \frac{q^2}{c} \left[\gamma^6 \{ \dot{\vec{\beta}}^2 - (\vec{\beta} \times \dot{\vec{\beta}})^2 \} \right]_{\text{ret}} \quad \text{relativistic Larmor Eqn}$$

non rel limit $|\vec{\beta}| \ll 1 \Rightarrow \gamma \approx 1 + O(\beta^2)$

$$P(t) \stackrel{|\beta| \ll 1}{\approx} \frac{2}{3} \frac{q^2}{c} \dot{\vec{\beta}}^2 + O(\beta^2)$$

$$\text{Particle physics } \gamma = \frac{E}{mc^2}, \quad \vec{\beta} = \frac{\vec{p}c}{E}, \quad E^2 = m^2 c^4 + \vec{p}^2 c^2$$

12/03
28/02

Electrodynamics

radiation by a moving point charge

rel Larmor eq. $P(t) = \frac{2}{3} \frac{q^2}{c} [\dot{\gamma}^2 \{ \dot{\vec{\beta}}^2 - (\dot{\vec{\beta}} \times \dot{\vec{\beta}})^2 \}]_{\text{rec}}$

Circular Motion

$|\vec{v}| = \omega \rho = \text{const}, \rho = \text{radius}$

$\vec{r}(t) = \rho (\cos \phi(t), \sin \phi(t), 0)^T$

$\vec{v}(t) \equiv \dot{\vec{r}}(t) = \rho \dot{\phi}(t) (-\sin \phi(t), \cos \phi(t), 0)^T \quad \dot{\phi} = \omega = \text{const}$

$\vec{v} = \dot{\vec{r}}(t) = \rho \omega^2 (-\cos \phi(t), -\sin \phi(t), 0)^T = -\omega^2 \vec{r}(t)$

$\Rightarrow \vec{v} \cdot \vec{v} = 0 \quad \underbrace{\dot{\vec{\beta}}^2 (1 - \beta^2)} = \dot{\vec{\beta}}^2 \gamma^{-2}$

$P = \frac{2}{3} \frac{q^2}{c} \gamma^6 (\dot{\vec{\beta}}^2 - (\dot{\vec{\beta}} \times \dot{\vec{\beta}})^2) = \frac{2}{3} \frac{q^2}{c} \gamma^6 \dot{\vec{\beta}}^2 (1 - \beta^2) = \frac{2}{3} \frac{q^2}{c} \gamma^4 \dot{\vec{\beta}}^2$

→ classical instability of atoms, as ~~as~~ electrons lose energy through radiation and spiral into the nucleus

so lifetime of atoms would be very short in classical physics

Collider Experiments

CERN-LHC: $2\pi\rho \approx 27\text{km}$

LEP: electron on positrons

protons on protons
Lead on Lead

● E_x electron $E = 100\text{GeV}$, $|\vec{\beta}| = c \frac{|\vec{p}|}{E} = \frac{\sqrt{E^2 - m_e^2 c^4}}{E} = \sqrt{1 - \left(\frac{m_e c^2}{E}\right)^2}$

$\rightarrow |\vec{\beta}| \approx 1 - \frac{1}{2} \left(\frac{m_e c^2}{E}\right)^2$

$|\vec{\beta}| \approx 1 - 0.13 \times 10^{-6}$

$\frac{m_e c^2}{E} = \frac{0.5\text{MeV}}{100 \times 10^3\text{MeV}} = 0.5 \times 10^{-5}$

$\Delta E = \frac{2\pi}{\omega} P \quad \gamma^4 \approx 1.6 \times 10^{21} \quad \gamma^{-15}$

$\uparrow = \frac{4\pi}{3} \frac{q^2}{\rho} \beta^3 \gamma^4 \quad \Delta E \approx \frac{4\pi}{3} \left(\frac{q^2}{m_e c^2}\right) m_e c^2 \frac{\gamma^4}{\rho}$

LHC $\rho \approx 5000\text{m}$ 0.5MeV

energy lost per revolution

$\Delta E \approx 6 \times 10^{-15} \cdot \frac{\gamma^4}{\rho} \text{MeV} \approx 2 \times 10^{-15+21-1} \frac{\text{GeV}}{\text{GeV}} \approx 2 \times 10^{-5} \text{GeV}$
2 GeV

13/03

Electrodynamics

radiation due to
a moving point charge

Radiation Loss

$$\Delta E = \frac{2\pi}{\omega} P = \frac{2\pi\rho}{c\rho} P = \frac{4\pi}{3} \frac{q^2}{\rho} \beta^3 \gamma^4 \propto \frac{\gamma^4}{\rho} \quad \gamma = \frac{E}{m_e c^2}$$

for corresponding Energies at same ΔE for protons vs electrons

$$m_p \approx 2000 m_e \quad \gamma_p^4 = \gamma_e^4$$

$$\frac{E_p}{m_p c^2} = \frac{E_e}{m_e c^2} \rightarrow \frac{E_p}{E_e} = \frac{m_p}{m_e} \approx 2000$$

Hw1 Q4: Lorentz boost in x direction of electric field

$$\vec{E}(\vec{x}, t) = \frac{q\vec{x}}{|\vec{x}|^3}, \quad \vec{B}(\vec{x}, t) = \vec{0}$$

$$E_x' = E_x$$

$$B_x' = B_x = 0$$

$$E_y' = \gamma(E_y - \beta B_z)$$

$$B_y' = \gamma(B_y + \beta E_z) = \gamma \beta E_z$$

$$E_z' = \gamma(E_z + \beta B_y)$$

$$B_z' = \gamma(B_z - \beta E_y) = -\gamma \beta E_y$$

$$\vec{E}'(\vec{x}', t') = \frac{q(\gamma(x' + \beta ct'), \gamma y', \gamma z')^T}{[\gamma^2(x' + \beta ct')^2 + y'^2 + z'^2]^{3/2}}$$

$$F'^{\mu\nu}(x') = \Lambda^\mu_\rho \Lambda^\nu_\sigma F^{\rho\sigma}(x) = \Lambda^\mu_\rho \Lambda^\nu_\sigma F^{\rho\sigma}(\Lambda^{-1}x')$$

$$E_x'(\vec{x}', t') = E_x(\vec{x}, t) = \frac{qx}{(x^2 + y^2 + z^2)^{3/2}}$$

$$= \frac{q\gamma(x' + \beta ct')}{(\gamma^2(x' + \beta ct')^2 + y'^2 + z'^2)^{3/2}}$$

$$ct' = \gamma(ct - \beta x)$$

$$x' = \gamma(x - \beta ct)$$

$$y' = y$$

$$z' = z$$

$$E_{rel}(\vec{x}, t) = \left[\frac{q(n^2 - \beta^2)}{R^2 \gamma^2 (1 - \vec{\beta} \cdot \vec{n})^3} \right]_{rel}$$

$$\vec{\beta} = (-\beta, 0, 0)^T$$

$$\vec{r}_{(t)} = (-\beta ct, 0, 0)^T$$

Electrodynamics

radiation due to moving point charge

compare boosted electrostatic field to ret time formalism

$$\vec{E}(\vec{x}, t) = \frac{q(\gamma(x+\beta ct), y, z)}{[(\gamma x + \gamma \beta ct)^2 + y^2 + z^2]^{3/2}} \leftarrow \text{electrostatic field of a point charge boosted in } x\text{-direction}$$

$$\vec{E}(\vec{x}, t) = \left[\frac{q(\vec{R} - \vec{\beta})}{R^3 \gamma^2 (1 - \vec{\beta} \cdot \hat{n})^3} \right]_{\text{ret}} \leftarrow \text{velocity field}$$

$$\vec{r}(t) = (-\beta ct, 0, 0)^T$$

$$ct_{\text{ret}} = ct - R$$

$$\vec{x} - \vec{r}(t_{\text{ret}}) = (x + \beta ct_{\text{ret}}, y, z)^T \quad c(t - t_{\text{ret}}) = |\vec{x} - \vec{r}(t_{\text{ret}})|$$

$$R\hat{n} = \vec{x} - \vec{r}(t_{\text{ret}}) = (x + \beta ct - \beta R, y, z)^T$$

$$R^2 \hat{n}^2 = R^2 = (x + \beta ct - \beta R)^2 + y^2 + z^2 \quad \text{solve for } R$$

2 solns $R_{\pm} = \underbrace{-\gamma [x + \beta ct]}_{\text{Positive}} \pm \underbrace{\sqrt{(\gamma x + \gamma \beta ct)^2 + y^2 + z^2}}_{\text{Positive}}$ R is a modulus so $R \geq 0 \Rightarrow R \equiv R_+$

$$\begin{aligned} [\vec{n} - \vec{\beta}]_{\text{ret}} &= \frac{1}{R} [\vec{x} - \vec{r}(t_{\text{ret}}) - R\vec{\beta}] = \frac{1}{R} (x + \beta ct - \beta R + \beta y, z)^T \\ &= \frac{1}{R} \begin{pmatrix} x + \beta ct \\ y \\ z \end{pmatrix} \end{aligned}$$

R denominator of $\vec{E}_{\text{vel}} = \gamma^2 (R - \vec{\beta} \cdot \hat{n} R)^3$

$$[R\hat{\beta} \cdot \hat{n}]_{\text{ret}} = \vec{\beta} \cdot (\vec{x} - \vec{r}(t_{\text{ret}})) = \begin{pmatrix} -\beta \\ 0 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} x + \beta ct - \beta R \\ y \\ z \end{pmatrix} = -\beta (x + \beta ct - \beta R)$$

$$[R - R\hat{\beta} \cdot \hat{n}]_{\text{ret}} = R - [R\hat{\beta} \cdot \hat{n}]_{\text{ret}}$$

$$= R + \beta (x + \beta ct) - \beta^2 R = R\gamma^{-2} + \beta (x + \beta ct)$$

$$\begin{aligned} \rightarrow \frac{R_+}{\gamma^2} + \beta (x + \beta ct) &= \frac{1}{\gamma^2} (-\gamma \beta (x + \gamma \beta ct) + \gamma \sqrt{(\gamma x + \gamma \beta ct)^2 + y^2 + z^2} + \beta (x + \beta ct) \gamma^2) \\ &= \frac{\sqrt{(\gamma x + \gamma \beta ct)^2 + y^2 + z^2}}{\gamma} \end{aligned}$$

$$\rightarrow E(\vec{x}, t) = \left[\frac{q(\vec{R} - \vec{\beta})}{R^3 \gamma^2 (1 - \vec{\beta} \cdot \hat{n})^3} \right] = \frac{q(x + \beta ct, y, z)^T \gamma^2}{\sqrt{(\gamma x + \gamma \beta ct)^2 + y^2 + z^2}^3 \gamma^2} = \frac{q(\gamma(x + \beta ct), y, z)^T}{[(\gamma x + \gamma \beta ct)^2 + y^2 + z^2]^{3/2}}$$

Same as our Lorentz boost expression!

19/03

Electromagnetism dynamics

Electromag in matter

SI units:

Max Eqn: $\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$ $\vec{\nabla} \cdot \vec{B} = 0$
 in vacuum $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$ $\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \frac{\partial \vec{E}}{\partial t}$

in medium: $\vec{\nabla} \cdot \vec{D} = \rho_{free}$ $\vec{\nabla} \cdot \vec{B} = 0$ $\vec{D} = \epsilon_0 \vec{E} + \vec{P}$ electric replacement
 $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$ $\vec{\nabla} \times \vec{H} = \vec{J}_{free} + \frac{\partial \vec{D}}{\partial t}$ $\vec{H} = \frac{1}{\mu_0} \vec{B} - \vec{M}$ magnetizing field or auxiliary field

magnetization

$$\vec{M} = \chi_m \vec{H}$$

In linear media $\vec{P} = \epsilon_0 \chi_e \vec{E}$
 polarization

$$\Rightarrow \vec{D} = \epsilon_0 (1 + \chi_e) \vec{E}, \vec{H} = \frac{1}{\mu_0} \vec{B}$$

$$\Rightarrow \vec{H} = \frac{1}{\mu_0 (1 + \chi_m)} \vec{B} = \frac{1}{\mu} \vec{B}$$

 χ_{em} electric/magnetic susceptibility

 μ_0, μ and $\mu_r \equiv \frac{\mu}{\mu_0}$ permeability of vacuum, medium and relative permeability

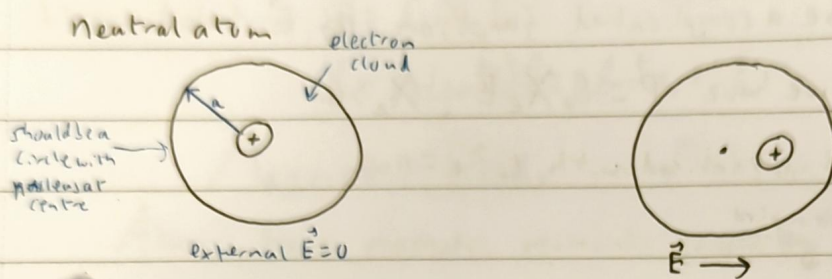
 ϵ_0, ϵ and $\epsilon_r \equiv \frac{\epsilon}{\epsilon_0}$ permittivity of - - - -

20/03

Electrodynamics

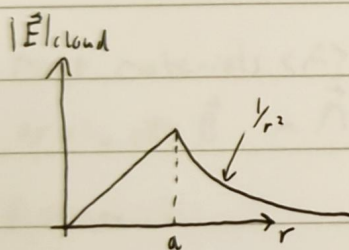
Electromag in matter

Model for atomic polarizability

Electrostatics: sphere (radius a) with uniform charge density (ρ_{cloud})

$$\vec{E}_{\text{cloud}}(\vec{x}) = \begin{cases} \frac{1}{4\pi\epsilon_0} \frac{q}{a^3} \vec{x}, & \text{if } r = |\vec{x}| < a \\ \frac{q}{4\pi\epsilon_0 |\vec{x}|^2}, & \text{if } r > a \end{cases}$$

$$\epsilon = \frac{\frac{4\pi}{3} a^3 \rho_{\text{cloud}}}{V_{\text{sphere}}}$$



now apply external field. The centre of the distribution is displaced wrt the nucleus

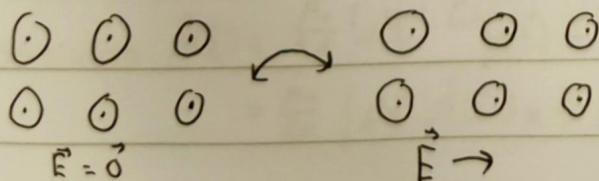
$$\text{unt. l } \vec{E}_{\text{ext}} + \vec{E}_{\text{cloud}}(\vec{x}_0) = 0 \Rightarrow \vec{x}_0 = -\frac{4\pi\epsilon_0 a^3}{q} \vec{E}_{\text{ext}}$$

 $\vec{E}_{\text{ext}}$ from now on

⇒ This displacement generates an e-dipole moment $\vec{d} = -q\vec{x}_0 = \underbrace{4\pi\epsilon_0 a^3}_{\alpha} \vec{E}_{\text{ext}}$

$$\vec{d} = \alpha \vec{E} \quad \alpha \equiv \text{atomic polarizability}$$

Material made up of a lattice of atoms



$$\vec{P} = \text{polarization} = \frac{\text{averaged } \vec{d}}{\text{un. r. volume}}$$

$$\vec{P} = n \cdot \vec{d} = n \alpha \vec{E}$$

↑
dens. of matter

20/19

Electrodynamics

E-mag in matter

Polarizability

in general, $\vec{P}$ can be a complicated function of $\vec{E}$ in linear dielectrics we have $\vec{P} = \epsilon_0 \chi_e \vec{E}$, $\chi_e > 0$ in the simple model this is realised with $\epsilon_0 \chi_e = n\alpha$ E-dipole implied
for now

Electric potential of dipole moment

$$\phi(\vec{x}) = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \vec{x}}{r^3}, \quad r = |\vec{x}|$$

integrate over all dipoles $\rightarrow \phi(x) = \frac{1}{4\pi\epsilon_0} \int d^3x' \frac{\vec{p}(\vec{x}') \cdot (\vec{x} - \vec{x}')}{|\vec{x} - \vec{x}'|^3}$ need to IBP

$$\vec{\nabla}' \frac{1}{|\vec{x} - \vec{x}'|} = \frac{\partial}{\partial x_k} \left(\sum_{i=1}^3 (x_i - x'_i)^2 \right)^{-1/2} = -\frac{1}{2} \left(\sum_{i=1}^3 (x_i - x'_i)^2 \right)^{-3/2} (-2)(x_k - x'_k)$$

$$= \frac{x_k - x'_k}{|\vec{x} - \vec{x}'|^3}$$

$$\Rightarrow \vec{\nabla}' \frac{1}{|\vec{x} - \vec{x}'|} = \frac{\vec{x} - \vec{x}'}{|\vec{x} - \vec{x}'|^3}$$

$$\rightarrow \phi(x) = \frac{1}{4\pi\epsilon_0} \int d^3x' \vec{p}(\vec{x}') \cdot \vec{\nabla}' \frac{1}{|\vec{x} - \vec{x}'|}$$

$$\text{IBP} = \frac{1}{4\pi\epsilon_0} \left(\underbrace{\int_{\partial V} d\vec{S} \cdot \frac{\vec{p}(\vec{x}')}{|\vec{x} - \vec{x}'|}}_{\substack{\text{Potential for} \\ \text{surface charge} \\ \text{density on } \partial V \\ \sigma_{\text{bound}}^{(2)} = \vec{p} \cdot \vec{n} \\ \uparrow \text{outward normal}}} - \underbrace{\int d^3x' \vec{\nabla}' \frac{\vec{p}(\vec{x}')}{|\vec{x} - \vec{x}'|}}_{\text{Potential for } \rho_{\text{bound}}(\vec{x}) \equiv -\vec{\nabla} \cdot \vec{p}(\vec{x})} \right)$$

Gauss law $\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$ always validElectric displacement $\vec{D}$ charges moving freely due to polarizationin matter, we can split $\rho = \rho_{\text{free}} + \rho_{\text{bound}}$

$$\text{rewrite } \vec{\nabla} \cdot \vec{E} = \frac{\rho_{\text{free}} + \rho_{\text{bound}}}{\epsilon_0} = \frac{1}{\epsilon_0} (\rho_{\text{free}} + \vec{\nabla} \cdot \vec{P})$$

$$\vec{D} \equiv \epsilon_0 \vec{E} + \vec{P} \Rightarrow \vec{\nabla} \cdot \vec{D} = \epsilon_0 \vec{\nabla} \cdot \vec{E} + \vec{\nabla} \cdot \vec{P} = \rho_{\text{free}}$$

20/03

Electrodynamics

E-M in matter

Linear Dielectrics: $\vec{D} = \epsilon_0 \chi_e \vec{E}$ $\epsilon_r = 1 + \chi_e = \text{relative permittivity}$

$$\vec{D} = \epsilon_0 (1 + \chi_e) \vec{E} = \epsilon \vec{E} = \epsilon_0 \epsilon_r \vec{E}$$

e.g. $\epsilon_r \approx 1$ for glasses $\epsilon_r \approx 80$ for water

Magnetic Fields in Matter

Atoms have magnetic moments, caused by

- electrons with angular momentum $\rightarrow$ magnetic dipole moment
- electron spins (QM origin)

magnetization

$$\vec{M}(\vec{x}) = n \langle \vec{m} \rangle$$

number density of atoms

average magnetic moment per atom

$$\langle \vec{m} \rangle = \frac{1}{N} \sum_{i=1}^N \vec{m}(\vec{x}_i)$$

Most materials $\langle \vec{m} \rangle = 0$ if $\vec{B} = 0$

magnetic susceptibility

$$\text{apply ext } \vec{B} \rightarrow \vec{M} \propto \vec{B}, \quad \vec{M} = \frac{1}{\mu_0} \frac{\chi_m}{1 + \chi_m} \vec{B}$$

distinguish:

- diamagnetic materials $-\chi_m < 0$ e.g. copper, gold, water, $\vec{M}$ antiparallel to $\vec{B}$
- Paramagnetic $\chi_m > 0$ e.g. aluminium, tungsten
- ferromagnetic $\rightarrow \vec{M} \neq 0$ at $\vec{B} = 0$, e.g. iron (below Curie temp)

Magnetostatics (Griffiths Ch 5)

$$\vec{A}(\vec{x}) = \frac{\mu_0}{4\pi} \frac{\vec{m} \times \vec{x}}{r^3} \quad \text{for a magnetic dipole moment } \vec{m}$$

$$\text{generalise } \vec{A}(\vec{x}) = \frac{\mu_0}{4\pi} \int d^3x' \frac{\vec{m}(\vec{x}') \times (\vec{x} - \vec{x}')}{|\vec{x} - \vec{x}'|^3} \quad \text{let } \frac{\vec{x} - \vec{x}'}{|\vec{x} - \vec{x}'|^3} = \vec{\nabla}' \frac{1}{|\vec{x} - \vec{x}'|}$$

$$= \frac{\mu_0}{4\pi} \int d^3x' \vec{m}(\vec{x}') \times \vec{\nabla}' \frac{1}{|\vec{x} - \vec{x}'|}$$

21/03

Electrodynamics

E-M in matter

→ Contd

summation for
implied on
repeated indices
↓

$$\begin{aligned}
 \text{IBP} \Rightarrow A_K(\vec{x}) &= \frac{\mu_0}{4\pi} \int d^3x' \epsilon_{Kij} \vec{M}_i(\vec{x}') \partial_j' \frac{1}{|\vec{x}-\vec{x}'|} = \text{surface term} - \frac{\mu_0}{4\pi} \int d^3x' \epsilon_{Kij} \partial_j' \vec{M}_i(\vec{x}') \frac{1}{|\vec{x}-\vec{x}'|} \\
 &= \text{surface term} - \underbrace{\frac{\mu_0}{4\pi} \int d^3x' \epsilon_{Kij} (\partial_j' \vec{M}_i(\vec{x}')) \frac{1}{|\vec{x}-\vec{x}'|}}_{\text{neglect surface}} \\
 &= \underbrace{\frac{\mu_0}{4\pi} \int d^3x' \frac{(\vec{\nabla} \times \vec{M})_K(\vec{x}')}{|\vec{x}-\vec{x}'|}}_{\text{potential of a current density}} d^3x'
 \end{aligned}$$

Biot-Savart: $\vec{A}(\vec{x}) = \frac{\mu_0}{4\pi} \int d^3x' \frac{\vec{j}(\vec{x}')}{|\vec{x}-\vec{x}'|}$

define $\vec{j}_{\text{bound}} = \vec{\nabla} \times \vec{M} \Rightarrow \vec{\nabla} \cdot \vec{j}_{\text{bound}} = 0$

$$\sum_{Kij} \epsilon_{Kij} \partial_i M_j = (\vec{\nabla} \times \vec{M})_K$$

$$\sum_{K, i, j} \epsilon_{Kij} \partial_K \partial_i M_j = \vec{\nabla} \cdot (\vec{\nabla} \times \vec{M}) = 0$$

Split $\vec{j} = \vec{j}_{\text{free}} + \vec{j}_{\text{bound}} = \vec{j}_{\text{free}} + \vec{\nabla} \times \vec{M}$

Ampère's law: $\vec{\nabla} \times \vec{B} = \mu_0 \vec{j} = \mu_0 (\vec{j}_{\text{free}} + \vec{\nabla} \times \vec{M})$

define $\vec{H} = \frac{1}{\mu_0} \vec{B} - \vec{M}$ magnetizing/auxiliary field

$$\vec{\nabla} \times \vec{H} = \vec{j}_{\text{free}}$$

Paramagnetic materials $\vec{B} = \mu \vec{H}$, $\vec{H} = \frac{1}{\mu} \vec{B} = \frac{1}{\mu_0} \vec{B} - \vec{M} = \frac{\vec{B}}{\mu_0} - \frac{1}{\mu_0} \frac{\chi_m}{1+\chi_m} \vec{B}$
 $= \frac{1}{\mu_0(1+\chi_m)} \vec{B}$

$$\Rightarrow \mu = \mu_0(1+\chi_m) = \mu_0 \mu_r$$

$$\vec{H} = \frac{1}{\mu} \vec{B} = \frac{\mu_0}{\mu} (\vec{H} + \vec{M}) \Rightarrow \vec{M} = \left(\frac{\mu}{\mu_0} - 1\right) \vec{H} = \chi_m \vec{H} \rightarrow \vec{M} = \chi_m \vec{H}$$

So now what about time dependence?

Use current conservation $\vec{\nabla} \cdot \vec{j}_{\text{bound}} + \frac{\partial \rho_{\text{bound}}}{\partial t} = 0$

021/03

Electrodynamics

E-M in Mater

Time dependant Magnetic

recall $\rho_{\text{bound}} = -\vec{\nabla} \cdot \vec{P}$

$$\vec{\nabla} \cdot \vec{J}_{\text{bound}} - \frac{\partial}{\partial t} \vec{\nabla} \cdot \vec{P} = 0 \Rightarrow \vec{J}_{\text{bound}} = \vec{\nabla} \times \vec{M} + \frac{\partial \vec{P}}{\partial t}$$

maxwell eqn from extension of Amperes law

$$\vec{\nabla} \times \vec{B} - \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} = \mu_0 \vec{J} = \mu_0 (\vec{J}_{\text{free}} + \vec{J}_{\text{bound}}) = \mu_0 \left(\vec{J}_{\text{free}} + \vec{\nabla} \times \vec{M} + \frac{\partial \vec{P}}{\partial t} \right)$$

$$\vec{\nabla} \times (\underbrace{\vec{B} - \mu_0 \vec{M}}_{\mu_0 \vec{H}}) - \mu_0 \epsilon_0 \frac{\partial}{\partial t} \left(\underbrace{\vec{E} + \frac{1}{\epsilon_0} \vec{P}}_{\frac{1}{\epsilon_0} \vec{D}} \right) = \mu_0 \vec{J}_{\text{free}}$$

$$\Rightarrow \vec{\nabla} \times \vec{H} - \frac{\partial}{\partial t} \vec{D} = \vec{J}_{\text{free}}$$

remaining Maxwells: $\vec{\nabla} \cdot \vec{D} = \rho_{\text{free}}$

$$\vec{\nabla} \cdot \vec{B} = 0, \quad \vec{\nabla} \times \vec{E} + \frac{\partial \vec{P}}{\partial t} = 0$$

Moving point charge

Paul Haskins

March 12, 2025

Rel Lamor eq

$$P(t) = \frac{2}{3} \frac{q^2}{c} [\gamma^6 (\beta^{\cdot 2} - (\beta \times \beta^{\cdot})^2)]_{ret} \quad (1)$$

non-rel limit ie $\beta \ll 1$

$$P(t) \approx \frac{2}{3} \frac{q^2}{c} (\beta^{\cdot 2})_{ret} + O(\beta^2) \quad (2)$$

$$(\gamma^2)^2 = \frac{1}{(1 - \beta^2)^2} \approx 1 + 2\beta^2 + \dots \quad (3)$$

Circular motion

$|\vec{v}|w\rho = \text{const}$, ρ = radius.

$$\vec{r}(t) = \rho (\cos \phi(t) \sin \phi(t), 0)^T \quad (4)$$

$$\vec{v}(t) = \dot{\vec{r}}(t) = \rho \dot{\phi}(t) (-\sin \phi(t) \sin \phi(t), 0)^T \quad (5)$$

$$\ddot{\vec{r}}(t) = \rho \omega^2(t) (\cos \phi(t) \sin \phi(t), 0)^T = -\omega^2 \vec{r}(t) \quad (6)$$

with $\omega = \dot{\phi}$. Now we can look at the power

$$P = \frac{2}{3} \frac{q^2}{c} \gamma^6 (\beta^{\cdot 2} - (\beta \times \beta^{\cdot})^2) \quad (7)$$

if we have $\beta \perp \beta^{\cdot}$

$$= \frac{2}{3} \frac{q^2}{c} \gamma^6 (\beta^{\cdot 2} - \beta^2 \beta^{\cdot 2}) = \frac{2}{3} \frac{q^2}{c} \gamma^4 \beta^{\cdot 2} \quad (8)$$

This lead to instability of atoms in the classical understanding as electrons lose energy through radiation and spiral into the nucleus.

0.0.1 (Collider Experiments e.g LHC)

for the case of LHC $2\pi\rho \approx 27$ km. Proton on proton action ;), Pb-Pb collisions.

LEP electron on positron action

Example

Electron $E = 100$ GeV

$$\vec{\beta} = \frac{c\vec{p}}{E} = \frac{\sqrt{E^2 - m_e^2 c^4}}{E} = \sqrt{1 - \left(\frac{m_e c^2}{E}\right)^2} \approx \quad (9)$$

$$\frac{m_e c^2}{E} = \frac{0.5 \text{ MeV}}{100 \times 10^3 \text{ MeV}} = 0.5 \times 10^{-5} \implies \beta \approx 1 - 0.13 \times 10^{-10} \quad (10)$$

$$\gamma = \frac{E}{m_e c^2} \approx 2 \times 10^5 \implies \gamma^4 = 1.6 \times 10^{21} \quad (11)$$

Energy loss for electron $\beta \approx 1$

$$\Delta E \approx \frac{4\pi}{3} \left(\frac{q^2}{m_e c^2} \right) m_e c^2 \frac{\gamma^4}{\rho} \quad (12)$$

where the thing in brackets is 2×10^{-15} electron radius. And the $m_e c^2$ is 0.5 MeV

In the case of the LHC.

$$\rho \approx 5000m, \Delta E \approx 6 \times 10^{-15} \frac{\gamma^4}{(\frac{\rho}{m}) \text{ MeV}} = 2 \times 10^{-15+21-3-3} \text{ GeV} = 2 \text{ GeV}.$$

To keep ΔE small while increasing the energy on needs to increase $\rho \propto \gamma^4 \propto E^4$.

Example to double the energy $\rho \rightarrow 2^4 * \rho$.

Since $\gamma = \frac{E}{m c^2}$ particles with larger mass radiate less at the same energy.

$$\frac{\Delta E_{\text{elec}}}{\Delta E_{\text{protons}}} = \left(\frac{\gamma_{el}}{\gamma_{el}} \right)^2 \propto (2000)^4 = 16 \times 10^{12} = 1.6 \times 10^{13} \quad (13)$$

Moving charged particle

Paul Haskins

March 13, 2025

Radiation loss

$$\Delta E = \frac{2\pi}{\omega} P = \frac{2\pi\rho}{c\beta} P = \frac{4\pi}{3} \frac{q^2}{\rho} \beta^3 \gamma^4 \propto \frac{\gamma^4}{\rho} \quad (1)$$

1 period = 1 revolution $\gamma = \frac{E}{mc^2}$

Ask for corresponding energies at some ΔE for protons vs electrons.

$$m_p \approx 2000 m_e \quad (2)$$

$$\text{same } \Delta E_p = \Delta E_{el} \implies \gamma_p^4 = \gamma_{el}^4$$

$$\frac{E_p}{m_p c^2} = \frac{E_{el}}{m_{el} c^2} \quad (3)$$

LEP: $E_{el} \approx 50 \text{ GeV}$ and LHC $E_p \approx 100 \text{ TeV}$. LHC currently $E_p \approx 13 \text{ TeV}$.

Limiting factor not radiator loss but the magnetic field required to keep protons and their circular trajectory.

0.0.1 (Lorentz boost in x-direction of electrostatic field HW1,Q4)

$$\vec{E}(\vec{x}, t) = \frac{q\vec{x}}{|\vec{x}|^3}, \vec{B}(\vec{x}, t) = 0 \quad (4)$$

$$E'_x = E_x \quad (5)$$

$$E'_y = \gamma(E_y - \beta B_z) = \gamma E_y \quad (6)$$

$$E'_z = \gamma(E_z + \beta B_y) = \gamma E_z \quad (7)$$

$$B'_x = B_x = 0 \quad (8)$$

$$E'_y = \gamma(B_y + \beta E_z) = \gamma \beta E_z \quad (9)$$

$$E'_z = \gamma(B_z - \beta E_y) = -\gamma \beta E_y \quad (10)$$

Lorentz covariant combination

$$\vec{E} \cdot \vec{B}, \vec{E}^2 - \vec{B}^2 \quad (11)$$

These can be checked see hw

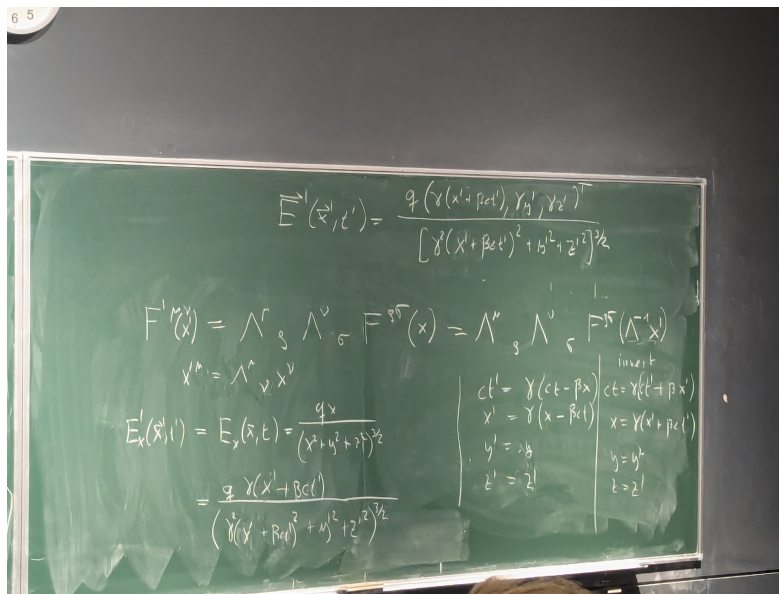


Figure 1: 2.1.jpg

$$E_{ret}(\vec{x}, t) = \left[\frac{q(\vec{n} \cdot \vec{\beta})}{R^2 \gamma^2 (1 - \beta \cdot \vec{n})^3} \right]_{ret} \quad (12)$$

which is a velocity field.

This should match the expression in the figure above.

$$\vec{\beta} = (-\beta, 0, 0)^T \quad (13)$$

$$\vec{r}(t) = (-\beta ct, 0, 0)^T \quad (14)$$

Trajectory of point charge.

Moving particle

Paul Haskins

March 19, 2025

$$\vec{E}(x, t) = \frac{q (\gamma (x + \beta ct), \gamma y, \gamma z)^T}{[(\gamma x + \gamma \beta ct)^2 + y^2 + z^2]^{\frac{3}{2}}} \quad (1)$$

$$(2)$$

Electrostatic field of point charge boosted in the x direction and compare to the velocity field in the retarded time formalism

$$\vec{E}(\vec{x}, t) = \left[\frac{q (\vec{n} - \vec{\beta})}{R^2 \gamma^2 (1 - \vec{\beta} \cdot \vec{n})^3} \right]_{kt} \quad (3)$$

Trajectory of a point charge

$$\vec{r}(t) = (-\beta ct, 0, 0)^T \quad (4)$$

$$\vec{x} - \vec{r}(t_{ret}) = (x + \beta ct_{ret}, y, z)^T = R \vec{n}, \vec{n}^2 = 1 \quad (5)$$

with

$$c(t - t_{ret}) = R = |\vec{x} - \vec{r}(t_{ret})| \quad (6)$$

$$\implies ct_{ret} = ct - R \quad (7)$$

$$R \vec{n} = \vec{x} - \vec{r}(t_{ret}) = (x + \beta ct - \beta R, y, z)^T \quad (8)$$

$$R^2 \vec{n}^2 = R^2 = (x + \beta ct - \beta R)^2 + y^2 + z^2 \quad (9)$$

Solve the quadratic for R. We get the solutions

$$R_{\pm} = -\gamma \left[\beta (\gamma x + \gamma \beta ct) \mp \sqrt{(\gamma x + \gamma \beta ct)^2 + y^2 + z^2} \right] \quad (10)$$

Where R is the modulus of a vector and therefore positive. We assume $\beta (\gamma x + \gamma \beta ct) \geq 0$. $\implies R_- < 0$ keep $R = R_+$

$$\left[\vec{n} - \vec{\beta} \right]_{ret} = \frac{1}{R} \left[(\vec{x} - \vec{r}(t_{ret})) - R \vec{\beta} \right] \quad (11)$$

$$= \frac{1}{R} [x + \beta ct - \beta R + R \beta, y, z]^T \quad (12)$$

$$= \frac{1}{R} \begin{pmatrix} x + \beta ct \\ y \\ z \end{pmatrix} \quad (13)$$

We have R times the denomination of $\vec{E}_{ret} = \gamma^2 (R - R \vec{\beta} \cdot \vec{n})^3$.

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$$\left[R - R \vec{\beta} \cdot \vec{n} \right]_{ret} = R - [R \beta n]_{ret} = R + \beta (x + \beta ct) - \beta^2 R = R \gamma^{-2} + \beta (x + \beta ct) \quad (14)$$

$$\frac{R_+}{\gamma^2} + \beta (x + \beta ct) = \frac{1}{\gamma^2} [-\gamma \beta (\gamma x + \gamma \beta ct) + \gamma \sqrt{\dots} + \beta (x + \beta ct) \gamma^2] = \frac{\sqrt{\dots}}{\gamma} \quad (15)$$

$$\left[\frac{q (n - \vec{\beta})}{R^2 \gamma^2 (n - \beta \cdot n)^3} \right]_{ret} = \frac{q (x + \beta ct, y, z)^T}{\sqrt{\dots}} \gamma^3 \quad (16)$$

$$= \frac{q (x + \beta ct, y, z)^T}{\gamma^2 (\gamma^2 [x + \beta ct]^2 + y^2 + z^2)^{\frac{3}{2}} \gamma^{-3}} \quad (17)$$

$$= \frac{\gamma q (x + \beta ct, y, z)^T}{(\gamma^2 (x + \beta ct)^2 + y^2 + z^2)^{\frac{3}{2}}} \quad (18)$$

Electromagnetism in matter (SI)

Maxwell equations in a vacuum.

$$\vec{\nabla} \vec{E} = \frac{\rho}{\epsilon_0} \quad (19)$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad (20)$$

$$\vec{\nabla} \cdot \vec{B} = 0 \quad (21)$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{j} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \quad (22)$$

in the case where we have a medium

$$\nabla \cdot \vec{D} = \rho_{free} \quad (23)$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad (24)$$

$$\vec{\nabla} \cdot \vec{B} = 0 \quad (25)$$

$$\vec{\nabla} \times \vec{H} = \vec{j}_{free} + \frac{\partial \vec{D}}{\partial t} \quad (26)$$

$$(27)$$

and we have the electric displacement

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P} \quad (28)$$

$$\vec{H} = \frac{1}{\mu_0} \vec{B} - \vec{M} \quad (29)$$

auxiliary field (griffiths) or magnetising field.

In linear media.

$$\vec{P} = \epsilon_0 \chi_e \vec{E} \quad (30)$$

$$\vec{M} = \chi_m \vec{H} \quad (31)$$

$$\implies \vec{D} = \epsilon_0 (1 + \chi_e) \vec{E} \quad (32)$$

we define $\epsilon = \epsilon_0 (1 + \chi_e)$

$$\vec{H} (1 + \chi_m) = \frac{1}{\mu_0} \vec{B} \quad (33)$$

$$\vec{H} = \frac{1}{\mu_0 (1 + \chi_m)} \vec{B} = \frac{1}{\mu} \vec{B} \quad (34)$$

Model for atomic polarize ability

Paul Haskins

March 20, 2025

In electrostatics we model atoms as a sphere (radius a) with a uniform charge density.

$$\vec{E}_{cloud}(x) = \begin{cases} \frac{1}{4\pi\epsilon_0} \frac{q}{a^3} \vec{x} & \text{if } r = |x| < a \\ \frac{q}{4\pi\epsilon_0 |x|^2} & \text{if } r > a \end{cases} \quad (1)$$

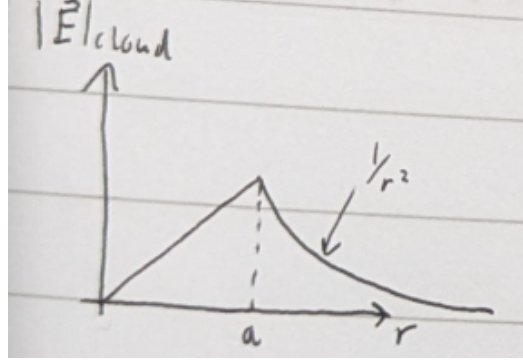


Figure 1: 4.1.png

Now if we apply an electric field. The centre of the distribution is displaced wrt the nucleus until

$$\vec{E}_{ext} + \vec{E}_{cloud}(x_0) = 0 \implies x_0 = -\frac{4\pi\epsilon_0 a^3}{q} E_{ext} \quad (2)$$

This displacement generates an electric dipole moment $\vec{d} = q\vec{x}_0 = 4\pi\epsilon_0 a^3 \vec{E}$. From now on we will call $\vec{E}_{ext} = \vec{E}$. and we call $\alpha = 4\pi\epsilon_0 a^3$. where α is the atomic polarizability.

If we consider a material made up of a lattice of atoms. The polarization $\vec{p}$

$$\vec{p} = \frac{\text{averaged } \vec{d}}{\text{unit volume}} \quad (3)$$

$$\vec{p} = n \cdot \vec{d} = n\alpha \vec{E} \quad (4)$$

where n is the density of the matter.

In general $\vec{P}$ can be a complicated function of $\vec{E}$ in linear dielectrics we have $\vec{P} = \epsilon_0 \chi_e \vec{E}$ with $\chi_e > 0$. In this simple model this is realised with $\epsilon_0 \chi_e = n\alpha$.

Electric potential of a dipole moment.

$$\phi(\vec{x}) = \frac{1}{4\pi\epsilon_0} \frac{\vec{d} \cdot \vec{x}}{r^3}, r = |\vec{x}| \quad (5)$$

If we integrate over all dipoles *need IBP*

$$\phi(\vec{x}) = \frac{1}{4\pi\epsilon_0} \int_V d^3\vec{x}' \frac{\vec{p}(\vec{x}') \cdot (\vec{x} - \vec{x}')}{|\vec{x} - \vec{x}'|^3} \quad (6)$$

$$\vec{\nabla}' \frac{1}{|\vec{x} - \vec{x}'|} : \frac{\partial}{\partial x'_k} \left(\sum_{i=1}^3 (x_i - x'_i)^2 \right)^{-\frac{1}{2}} = -\frac{1}{2} \left(\sum_i (x_i - x'_i)^2 \right)^{-\frac{3}{2}} (-2)(x_k - x'_k) \quad (7)$$

$$= \frac{x_k - x'_k}{|\vec{x} - \vec{x}'|^3} \quad (8)$$

$$\Rightarrow \vec{\nabla}' \frac{1}{|\vec{x} - \vec{x}'|} = \frac{\vec{x} - \vec{x}'}{|\vec{x} - \vec{x}'|^3} \quad (9)$$

$$\Rightarrow \phi(x) = \frac{1}{4\pi\epsilon_0} \int_V d^3\vec{x}' \vec{P}(\vec{x}') \cdot \vec{\nabla}' \frac{1}{|\vec{x} - \vec{x}'|} \quad (10)$$

$$\text{IBP} \Rightarrow \frac{1}{4\pi\epsilon_0} \left[\int_{\partial V} d\vec{S} \cdot \frac{\vec{p}(\vec{x}')}{|\vec{x} - \vec{x}'|} - \int_V d^3\vec{x}' \frac{\vec{\nabla}' \cdot \vec{P}(\vec{x}')}{|\vec{x} - \vec{x}'|} \right] \quad (11)$$

Where the first term is the potential for the surface charge density on ∂V , $\sigma_{\text{bound}}(\vec{x}) = \beta \cdot \vec{n}$, with $\vec{n}$ being the outward normal. The second term including neg is the potential for $P_{\text{bound}}(\vec{x}) = -\vec{\nabla}' \cdot \vec{P}(\vec{x}')$.

Gauss's law $\nabla \cdot E = \frac{\rho}{\epsilon}$ always valid bro.

Electric displacement = $\vec{D}$

In matter we can split $\rho = \rho_{\text{free}} + \rho_{\text{bound}}$ where ρ_{free} charges move freely, ρ_{bound} is due to the polarisation.

Rewrite

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho_{\text{free}} + \rho_{\text{bound}}}{\epsilon_0} = \frac{1}{\epsilon_0} (\rho_{\text{free}} + \nabla \cdot P) \quad (12)$$

$$\vec{D} \equiv \epsilon_0 \vec{E} + \vec{P} \Rightarrow \vec{D} = \epsilon_0 \nabla \cdot E + \nabla P = \rho_{\text{free}} \epsilon \quad (13)$$

In the case of linear dielectrics:

$$\vec{P} = \epsilon_0 \chi_e \vec{E} \quad (14)$$

$$\vec{D} = \epsilon_0 (1 + \chi_e) \vec{E} = \epsilon \vec{E} = \epsilon_0 \epsilon_r \vec{E} \quad (15)$$

where we define $\epsilon_r = 1 + \chi_e$ as the relative permittivity. $\epsilon_r \approx 1$ for gasses and $\epsilon_r \approx 80$ for water.

(Magnetic Fields in Matter)

Atoms have magnetic moments caused by:

electrons with angular momentum $\rightarrow$ magnetic dipole moment,

electron spins (QM)

$$M(x) = n \langle \vec{m} \rangle \quad (16)$$

$$\langle m \rangle = \frac{1}{N} \sum_{i=1}^N m(\vec{x}_i) \quad (17)$$

Where M is magnetization. n is the 'number density of atoms' and $\langle m \rangle$ is the average magnetic moment per atom.

Magnetic fields in matter

Paul Haskins

March 21, 2025

Either due to angular momentum or spin $\rightarrow$ magnetic moment

$$\vec{M} = n \langle \vec{m} \rangle \quad (1)$$

$$\langle \vec{m} \rangle = \frac{1}{N} \sum_{i=1}^N \vec{m}(\vec{x}_i) = \frac{\text{av mag moment}}{\text{atom}} \quad (2)$$

Most materials have $\langle \vec{m} \rangle = 0$ if $\vec{B} = 0$ no external magnetic field.

If we apply a magnetic field then

$$\vec{B} \propto \vec{M} \quad (3)$$

$$\vec{M} = \frac{1}{\mu_0} \frac{\chi_m}{1 + \chi_m} \vec{B} \quad (4)$$

$$(5)$$

where χ_m is magnetic susceptibility.

Dag magnetic materials $-1 < \chi_m < 0$ e.g copper, gold, water. $\vec{M}$ aligned to $\vec{B}$ but opposite directions.

Paramagnetic materials $\chi_m > 0$ e.g aluminum and tungsten.

Ferromagnetic materials $\vec{M} \neq 0$ at $\vec{B} = 0$ e.g iron, nickel. (below curie temperature)

While $\chi_e > 0$, χ_m can have either sign.

Assume we have a magnetic moment then we will incorporate it into our Maxwell equations.

Magneto statics (Griffiths chapter 5)

$$\vec{A}(\vec{x}) = \frac{\mu_0}{4\pi} \frac{\vec{m} \times \vec{x}}{r^3} \quad (6)$$

for a magnetic dipole $\vec{m}$. (magnet where we limit two points together).

We can generalize

$$A(\vec{x}) = \frac{\mu_0}{4\pi} \int_V d^3\vec{x}' \frac{\vec{M}(\vec{x}') \times (\vec{x} - \vec{x}')}{|\vec{x} - \vec{x}'|^3} \quad (7)$$

we can use

$$\frac{\vec{x} - \vec{x}'}{|\vec{x} - \vec{x}'|^3} = \vec{\nabla}' \frac{1}{|\vec{x} - \vec{x}'|} \quad (8)$$

to get

$$= \frac{\mu_0}{4\pi} \int_V d^3\vec{x} \vec{M}(\vec{x}) \times \vec{\nabla}' \frac{1}{|\vec{x} - \vec{x}'|} \quad (9)$$

IBP

$$A_k(\vec{x}) = \frac{\mu_0}{4\pi} \int_V d^3x' \sum_{ijk} \epsilon_{ijk} \vec{M}_i(\vec{x}') \partial'_j \frac{1}{|\vec{x} - \vec{x}'|} \quad (10)$$

$$= \text{surface term} - \frac{\mu_0}{4\pi} \sum_{ijk} \int_V \epsilon_{kij} \left(\partial'_j \vec{M}_i(\vec{x}') \right) \frac{1}{|\vec{x} - \vec{x}'|} \quad (11)$$

$$\approx -\frac{\mu_0}{4\pi} \int_V \sum_{ijk} (-\epsilon_{ijk}) \frac{\left(\partial'_j \vec{M}_i(\vec{x}') \right)}{|\vec{x} - \vec{x}'|} d^3\vec{x}' \quad (12)$$

$$= +\frac{\mu_0}{4\pi} \int \frac{\left(\vec{\nabla}' \times \vec{M} \right)_k(\vec{x}')}{|\vec{x} - \vec{x}'|} d^3\vec{x}' \quad (13)$$

Where we have dropped the surface term and we have a potential of a current density (Biot savart law).

$$\vec{A}(\vec{x}) = \frac{\mu_0}{4\pi} \int \frac{(\vec{\nabla}' \times \vec{M})_k(\vec{x}')}{|\vec{x} - \vec{x}'|} d^3\vec{x}' \quad (14)$$

define

$$\vec{j}_{bound} = \vec{\nabla} \times \vec{M} \implies \vec{\nabla} \cdot \vec{j}_{bound} = 0 \quad (15)$$

$$\epsilon_{kij} \partial_j M_j = (\vec{\nabla} \times \vec{M})_k \quad (16)$$

$$\epsilon_{kij} \partial_k \partial_i M_j = \vec{\nabla} \cdot (\vec{\nabla} \times \vec{M}) = 0 \quad (17)$$

vanishes as we contract antisymmetric with symmetric.

currents which satisfy this are called steady currents

Split

$$\vec{j} = \vec{j}_{free} + \vec{j}_{bound} = \vec{j}_{free} + \vec{\nabla} \times \vec{M} \quad (18)$$

Ampere's law

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{j} - \mu_0 (\vec{j}_{free} + \vec{\nabla} \times \vec{M}) \quad (19)$$

Define $\vec{H} = \frac{1}{\mu_0} \vec{B} - \vec{M}$ (magnetising/ auxiliary field). since cross products (rotations) are linear should be able to combine them.

$$\vec{\nabla} \times \vec{H} = \vec{j}_{free} \quad (20)$$

For para or diamagnetic materials we have

$$\vec{B} = \mu \vec{H} \quad (21)$$

$$\vec{H} = \frac{1}{\mu_0} \vec{B} - \vec{M} = \frac{\vec{B}}{\mu_0} \frac{\chi_m}{1 + \chi_m} \quad (22)$$

$$= \frac{1}{\mu_0 (1 + \chi_m)} \vec{B} \quad (23)$$

$$\implies \mu = \mu_0 (1 + \chi_m) = \mu_0 \mu_r \quad (24)$$

with $1 = \chi_m = \mu_r$.

$$\vec{H} = \frac{1}{\mu} \vec{B} = \frac{\mu_0}{\mu} (\vec{H} + \vec{M}) = \vec{M} = \left(\frac{\mu}{\mu_0} - 1 \right) \vec{H} = \chi_m \vec{H} \quad (25)$$

$$\vec{M} = \chi_m \vec{H} \quad (26)$$

So far we have talked about time independent systems.

For time dependent we will use current conservation $\vec{\nabla} \cdot \vec{j}_{bound} + \frac{\partial \rho_{bound}}{\partial t} = 0$
recall

$$\rho_{bound} = -\vec{\nabla} \cdot \vec{P} \quad (27)$$

$$\vec{\nabla} \cdot \vec{j}_{bound} - \frac{\partial}{\partial t} \vec{\nabla} \cdot \vec{P} = 0 \quad (28)$$

$$\implies \vec{j}_{bound} = \vec{\nabla} \times \vec{M} + \frac{\partial \vec{P}}{\partial t} \quad (29)$$

$$(30)$$

Maxwell equation from Ampere's law

$$\vec{\nabla} \times \vec{B} \mu_0 - \epsilon_0 \frac{\partial \vec{E}}{\partial t} = \mu_0 \vec{j} = \mu_0 \left(\vec{j}_{free} + \vec{j}_{bound} \right) = \mu_0 \left(\vec{j}_{free} + \vec{\nabla} \times \vec{M} + \frac{\partial \vec{P}}{\partial t} \right) \quad (31)$$

$$\vec{\nabla} \times \left(\vec{B} - \mu_0 \vec{M} \right) - \mu_0 \epsilon_0 \frac{\partial}{\partial t} \left(\vec{E} + \frac{1}{\epsilon_0} \vec{P} \right) = \mu_0 \vec{j}_{free} \quad (32)$$

with $\left(\vec{B} - \mu_0 \vec{M} \right) = \mu_0 \vec{H}$ and $\left(\vec{E} + \frac{1}{\epsilon_0} \vec{P} \right) = \frac{1}{\epsilon_0} \vec{D}$.

$$\vec{\nabla} \times \vec{H} - \frac{\partial \vec{D}}{\partial t} = \vec{j}_{free} \quad (33)$$

remaining equations

$$\vec{\nabla} \cdot \vec{D} = \rho_{free} \quad (34)$$

$$\vec{\nabla} \cdot \vec{B} = 0 \quad (35)$$

$$\vec{\nabla} \times \vec{E} + \frac{\partial \vec{B}}{\partial t} = 0 \quad (36)$$

Paul Haskins

March 26, 2025

$$\vec{\nabla} \times \vec{H} - \frac{\partial \vec{D}}{\partial t} = \vec{j}_{free} \quad (1)$$

$$\vec{\nabla} \cdot \vec{D} = \rho_{free} \quad (2)$$

$$\vec{\nabla} \times \vec{E} + \frac{\partial \vec{B}}{\partial t} = 0, \quad (3)$$

$$\vec{\nabla} \cdot \vec{B} = 0 \quad (4)$$

Focus on dielectric media, i.e. electric insulators that can be polarised

$$\rho_{free} = 0, \quad \vec{j}_{free} = 0 \quad (5)$$

In addition we assume linear media

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P} = \epsilon \vec{E} \quad (6)$$

$$\vec{H} = \frac{1}{\mu_0} - \vec{M} = \frac{1}{\mu} \vec{B} \quad (7)$$

Use these to replace $\vec{D}$ and $\vec{H}$

$$\frac{1}{\mu} \vec{\nabla} \times \vec{B} - \mu \epsilon \frac{\partial \vec{E}}{\partial t} = \vec{0}, \quad (8)$$

$$\epsilon \vec{\nabla} \cdot \vec{E} = 0 \quad (9)$$

$$\vec{\nabla} \times \vec{E} + \frac{\partial \vec{B}}{\partial t} = 0 \quad (10)$$

$$\vec{\nabla} \cdot \vec{B} = 0 \quad (11)$$

Differentiate eqn (8) wrt t

$$\implies \vec{\nabla} \times \frac{\partial \vec{B}}{\partial t} - \mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2} = \vec{0} \quad (12)$$

$$\frac{\partial \vec{B}}{\partial t} = -\vec{\nabla} \times \vec{E} \quad (13)$$

$$-\vec{\nabla} \times (\gamma \times \vec{E}) - \mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2} = 0 \quad (14)$$

$$-\gamma \times (\gamma \times \vec{E}) = -\vec{\nabla}^2 \vec{E} \text{ if } \vec{\nabla} \cdot \vec{E} = 0 \quad (15)$$

$$\mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2} - \vec{\nabla}^2 \cdot \vec{E} = 0 \quad (16)$$

We can do the same for $\vec{B}$ analogously we get

$$\mu \epsilon \frac{\partial^2 \vec{B}}{\partial t^2} - \vec{\nabla}^2 \cdot \vec{B} = 0 \quad (17)$$

These are the wave equation for the $\vec{E}$ and $\vec{B}$ fields velocity $v = \frac{1}{\sqrt{\mu \epsilon}}$. Here we have worked in XXX units while in the SI units we would easily identify the d Lambert operator.

Make an ansatz due to the fact that the $\vec{E}$ field is linear in our wave eqn

$$\vec{E}(\vec{x}, t) = \vec{E}_0 e^{i(\vec{k} \cdot \vec{x} - \omega t)} \quad (18)$$

These fields are complex, but physical fields must be real.

The physical $\vec{E}$, $\vec{B}$ fields correspond to $\Re \vec{E}$ and $\Re \vec{B}$.

In Griffith's $\vec{E}$ for complex fields $\vec{E}$ and $\vec{B}$ for physical fields.

Due to the linearity of the wave equation (and the MW eqs)

$$\Re \left[\left(\frac{1}{v^2} \frac{\partial^2}{\partial t^2} - \vec{\nabla}^2 \right) \vec{E}(\vec{x}, t) \right] = \left(\frac{1}{v^2} \frac{\partial^2}{\partial t^2} - \vec{\nabla}^2 \right) \Re \vec{E}(\vec{x}, t) \quad (19)$$

Complexified fields make the algebra easier. Observables are often the energy flux often quadratic in the fields.

WARNING

Quantities that are non-linear in the fields e.g.

$$\vec{S} \propto \vec{E} \times \vec{B} \equiv (\Re \vec{E}) \times (\Re \vec{B}) \neq \Re (\vec{E} \times \vec{B}) \quad (20)$$

$\exists$ a trick to get energy transport in plane wave (maybe on exam).

Now to use the ansatz

$$\frac{\partial \vec{E}}{\partial t} = -i\omega \vec{E}, \quad \frac{\partial \vec{E}}{\partial x_i} = ik_i \vec{E} \quad (21)$$

$$\frac{\partial^2}{\partial t^2} \vec{E} = (-i\omega)^2 \vec{E}, \quad \frac{\partial^2}{\partial x_i^2} \vec{E} = (ik_i)^2 \vec{E} \quad (22)$$

$$\left[\frac{1}{v^2} (-i\omega)^2 - (\vec{k})^2 \right] \vec{E} = 0 \quad (23)$$

$$= \frac{\omega^2}{v^2} + \vec{k}^2 = 0 \text{ or } \vec{k}^2 = \frac{\omega^2}{v^2} \quad (24)$$

Why a plane wave?

$$\vec{E}(\vec{x}, t) = \vec{E}_0 e^{i\phi(\vec{x}, t)} \quad (25)$$

phase $\phi(\vec{x}, t) = \vec{k} \cdot \vec{x} - \omega t$

A $t = t_0$ phase $\phi_0 = \phi(\vec{x}, t_0) = \vec{k} \cdot \vec{x} - \omega t_0$ keep fixed

Solutions $\vec{x}$ of $\phi(\vec{x}, t_0) = \phi_0$ all points for which $\vec{k} \cdot \vec{x} = \phi_0 + \omega t_0 = \text{const} = |\vec{k}| \cdot x_{||}$ (parallel to $\vec{k}$).

INCLUDE IDAGRMA XXX

$$\frac{\partial}{\partial t} \phi(\vec{x}, t) = |\vec{k}| \cdot \dot{x}_{||} - \omega = 0 \quad (26)$$

$$\implies \dot{x}_{||} = \frac{\omega}{|\vec{k}|} = v \quad (27)$$

$\implies$ Wave front moves with phase velocity in direction $\hat{k}$

Wavelength $e^{i\phi} = e^{i\phi + 2\pi m i}$ $m \in \mathbb{Z}$. if

$$(x_{||} + \Delta x_{||}) k - \omega t = x_{||} k - \omega t + 2\pi m \quad (28)$$

same values of $\vec{E}$ field. With $\Delta x_{||} = \frac{2\pi m}{k}$ and $\lambda = \frac{2\pi}{k}$ the minimal non-zero $\Delta x_{||}$.

$$\lambda = \frac{2\pi}{k} = \frac{2\pi}{\omega} \frac{\omega}{k} = Tv \quad (29)$$

6.1.jpg

Figure 1:

where T is the period. ν frequency given by

$$\frac{1}{T} = \nu = \frac{\omega}{2\pi} \quad (30)$$

Going back to Maxwell

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad (31)$$

$$i\vec{k} \times \vec{E} = i\omega \vec{B} \quad (32)$$

$$\vec{\nabla} \cdot \vec{B} = 0 \implies \vec{k} \cdot \vec{B} = 0 \quad (33)$$

$$\vec{\nabla} \cdot \vec{E} = 0 \implies \vec{k} \cdot \vec{E} = 0 \quad (34)$$

$$\vec{E}, \vec{B} \perp \vec{k} \quad (35)$$

transverse wave.

$$\vec{E} \cdot (\vec{k} \times \vec{E}) = \vec{e} \cdot \vec{B} = 0 \quad (36)$$

where $\text{LHS} = 0 \implies \vec{E} \perp \vec{B}$. Using right hand rule we can find perpendicular

Notation: $\hat{x}, \hat{y}, \hat{z}$, unit vectors in x, y, z direction.

choose $\frac{\vec{k}}{|\vec{k}|} = \hat{z}$, $k \equiv |\vec{k}|$

$$\vec{E}(\vec{x}, t) = \vec{E}_0 e^{i(kz - \omega t)}, \quad \vec{B}(\vec{x}, t) = \vec{B}_0 e^{i(kz - \omega t)} = \frac{(\vec{k} \times \vec{E})}{\omega} e^{i(kz - \omega t)}$$

$$\vec{E}_0 = (E_{0x}, E_{0y}, 0)^T, \quad E_{0x}, E_{0y} \in \mathbb{C}$$

convention: polarization defined by $\vec{E}$ -field

$$E_{0x} = |E_{0x}| e^{i\phi}, \quad E_{0y} = |E_{0y}| e^{i(\phi + \delta)}$$

$$\begin{aligned} \text{Physical field: } \text{Re } \vec{E}(\vec{x}, t) &= \text{Re} \left[(\hat{x} E_{0x} + \hat{y} E_{0y}) e^{i(kz - \omega t)} \right] \\ &= \hat{x} |E_{0x}| \cos(kz - \omega t + \phi) + \hat{y} |E_{0y}| \cos(kz - \omega t + \phi + \delta) \end{aligned}$$

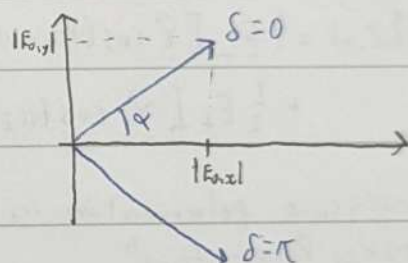
$$\cos(\delta + \alpha) = \cos \alpha \cos \delta - \sin \alpha \sin \delta$$

Distinguish cases

$$1) \delta = 0 \text{ or } \delta = \pi, \quad \cos(\alpha + \pi) = -\cos \alpha$$

$$\rightarrow \text{Re } \vec{E}(\vec{x}, t) = (\hat{x} |E_{0x}| \pm \hat{y} |E_{0y}|) \cos(kz - \omega t + \phi)$$

$\delta = \pi$
fixed vector, $\vec{x}, t$ independent



$\Rightarrow$ Linear polarization

$\rightarrow$ direction of pol. in xy plane

$$\tan \alpha = \pm \frac{|E_{0y}|}{|E_{0x}|}$$

$$\left. \begin{aligned} \hat{x} E_{0x} e^{i(kz - \omega t)} \\ \hat{y} E_{0y} e^{i(kz - \omega t)} \end{aligned} \right\} \begin{aligned} &\text{are 2} \\ &\text{linearly} \\ &\text{polarised waves} \end{aligned}$$

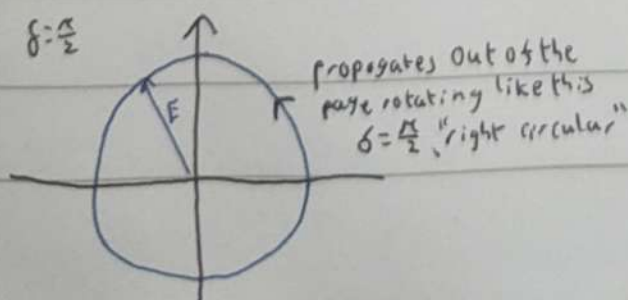
$\Rightarrow$ any plane wave can be thought of as a superposition of 2 linearly polarised waves

$$2) \delta = \pm \frac{\pi}{2}, \quad e^{\pm i\frac{\pi}{2}} = \pm i$$

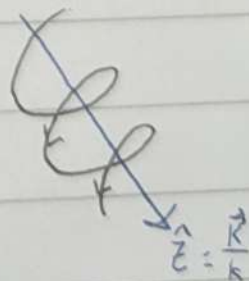
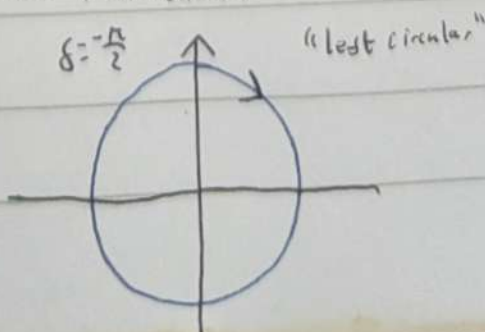
$$\text{Re } \vec{E}(\vec{x}, t) = \hat{x} |E_{0x}| \cos(kz - \omega t + \phi) \mp |E_{0y}| \sin(kz - \omega t + \phi)$$

if we have $|E_{0x}| = |E_{0y}| = E \Rightarrow$ circular polarization

$$\delta = \frac{\pi}{2}$$



$$\delta = -\frac{\pi}{2}$$



Polarization

Electrodynamics

27/03

if $\delta = \pm \frac{\pi}{2}$ but $|E_{0x}| \neq |E_{0y}|$

$$\rightarrow \left(\frac{\operatorname{Re}(E_x)}{|E_{0x}|} \right)^2 + \left(\frac{\operatorname{Re}(E_y)}{|E_{0y}|} \right)^2 = \cos^2(kz - \omega t + \phi) + \sin^2(kz - \omega t + \phi) = 1$$

makes an ellipse with axes $|E_{0x}|, |E_{0y}|$

"elliptic polarisation"

introduce vectors $\vec{u}_{\pm} = \frac{1}{\sqrt{2}}(\vec{x} \pm i\vec{y})$

$$\rightarrow \vec{x} = \frac{1}{\sqrt{2}}(\vec{u}_+ + \vec{u}_-), \quad \vec{y} = -\frac{i}{\sqrt{2}}(\vec{u}_+ - \vec{u}_-)$$

$$E_{0x} \pm iE_{0y} = E_{\pm} e^{i\gamma_{\pm}}$$

$$\rightarrow \vec{E}(\vec{x}, t) = \frac{1}{\sqrt{2}} \left[E_- \vec{u}_+ e^{i(kz - \omega t + \gamma_-)} + E_+ \vec{u}_- e^{i(kz - \omega t + \gamma_+)} \right] \equiv \vec{E}_0 e^{i(k\vec{x} - \omega t)}$$

$$\operatorname{Re} \vec{E}(\vec{x}, t) = \frac{1}{2} E_- \left[\vec{x} \cos(kz - \omega t + \gamma_-) - \vec{y} \sin(kz - \omega t + \gamma_-) \right] + \frac{1}{2} E_+ \left[\vec{x} \cos(kz - \omega t + \gamma_+) - \vec{y} \sin(kz - \omega t + \gamma_+) \right]$$

} Superposition of left & right circularly polarized waves

To measure polarization in real life we use "Stokes Parameters"

Energy transport in plane waves

Paul Haskins

March 28, 2025

Poynting vector :

$$\vec{s} = \vec{E} \times \vec{H} \quad (1)$$

if we assume linear media

$$= \frac{1}{\mu} \vec{E} \times \vec{B} \quad (2)$$

Energy density

$$u = \frac{1}{2} (\vec{H} \cdot \vec{B} + \vec{E} \cdot \vec{D}) = \frac{1}{2} \left(\frac{1}{\mu} \vec{B}^2 + \epsilon \vec{E}^2 \right) \quad (3)$$

quadratic in the physical fields.

To correctly compute $\vec{S}, u$ one computes

$$\vec{S} = \frac{1}{\mu} \Re \vec{E} \times \Re \vec{B} \neq \frac{1}{\mu} \Re (\vec{E} \times \vec{B}) \quad (4)$$

if $\vec{E}, \vec{B}$ and our complex fields

$\exists$ a shortcut via time averaging, visible light $\lambda = \mathcal{O}(100nm)$, in a vacuum $\lambda v = c$, $v = \frac{1}{T}$ with $T \equiv$ period

$$T = \frac{\lambda}{c} \approx \frac{10^{-7}m}{3 \cdot 10^8 \frac{m}{s}} = \frac{1}{3} \times 10^{-15}s \quad (5)$$

$\rightarrow$ shorter than the time resolution of most experiments.

observation

$$\begin{cases} a(t) = a_0 e^{-i\omega t} \\ b(t) = b_0 e^{-\omega t} \end{cases} \quad \text{otherwise} \quad (6)$$

$$(\Re a(t))(\Re b(t)) = \frac{1}{2} (a(t) + a^*(t)) \frac{1}{2} (b(t) + b^*(t)) \quad (7)$$

$$= \frac{1}{4} (a_0 b_0 e^{-2i\omega t} + a_0 b_0^* + a_0^* b_0 + a_0^* b_0^* e^{2i\omega t}) \quad (8)$$

We introduce the time average

$$\bar{f}(t) = \frac{1}{T} \int_t^{t+T} f(t') dt' \quad (9)$$

$$\frac{1}{T} \int_t^{t+T} dt' e^{\pm 2i\omega t'} = \frac{1}{T} \frac{1}{\pm 2i\omega} e^{\pm 2i\omega t'} \Big|_t^{t+T} \quad (10)$$

$$\implies T = \frac{1}{\nu} = \frac{2\pi}{\omega} \quad (11)$$

$$\overline{(\Re a(t))(\Re b(t))} = \frac{1}{4} (a_0 b_0^* + a_0^* b_0) = \frac{1}{2} (a_0 b_0^*) = \frac{1}{2} \Re (a(t) b^*(t)) \quad (12)$$

Rather than taking real parts of both we can take the conjugate of the second then cross them and multiply by a half. something along those lines.

$$\vec{s} = \frac{1}{2} \Re (\vec{E} \times \vec{H}^*) \quad (13)$$

$$\text{linear media} = \frac{1}{2\mu} \Re (\vec{E} \times \vec{B}^*) \quad (14)$$

$$\bar{u} = \frac{1}{4} \Re (H \cdot B^* + \vec{E} \cdot \vec{D}^*) = \frac{1}{4} \Re \left(\frac{1}{\mu} \vec{B} \cdot \vec{B}^* + \epsilon \vec{E} \cdot \vec{E}^* \right) \quad (15)$$

$$\vec{\nabla} = |\vec{B}|^2 = (\Re \vec{B})^2 + (\Im \vec{B})^2 \neq \Re |\vec{B}|^2 \quad (16)$$

Poynting vector

$$\vec{s} = \frac{1}{2\mu} \Re (\vec{E} \times \vec{B}^*) = \frac{1}{2\mu\omega} [\vec{E}_0 \times (\vec{k} \times \vec{E}_0)] \quad (17)$$

$$= \frac{1}{2\omega\mu} \Re (\vec{k} \vec{E}_0 \vec{E}_0^* - \vec{E}_0^* (\vec{E}_0 \cdot \vec{k})) = \frac{1}{2} \frac{k}{\omega} \frac{1}{\mu} \vec{E}_0 \vec{E}_0^* \cdot \vec{k} \quad (18)$$

$$= \frac{k}{\omega} \frac{1}{\mu\epsilon} \vec{u} \hat{k} = v \vec{u} \hat{k} \quad (19)$$

where $\frac{k}{\omega} = \frac{1}{v}$ and $\frac{1}{\mu\epsilon} = v^2$

$\vec{s}$ is an energy current density, energy density time velocity. units

$$\left[\frac{\vec{s}}{s} \right] = \frac{\text{energy}}{3 \text{ volume}} \cdot \text{velocity} = \frac{\text{energy}}{\text{area} \cdot \text{time}} = \frac{\text{power}}{\text{area}} = \frac{W}{m^2} \quad (20)$$

Radiation power

$$\frac{\vec{s}}{v} = \frac{\text{force}}{\text{area}} \quad (21)$$

Typical question calculate the time energy exactly or time average

(Reflection and transmission of EM waves at interfaces between linear dielectric media e.g water, air, glass ...)

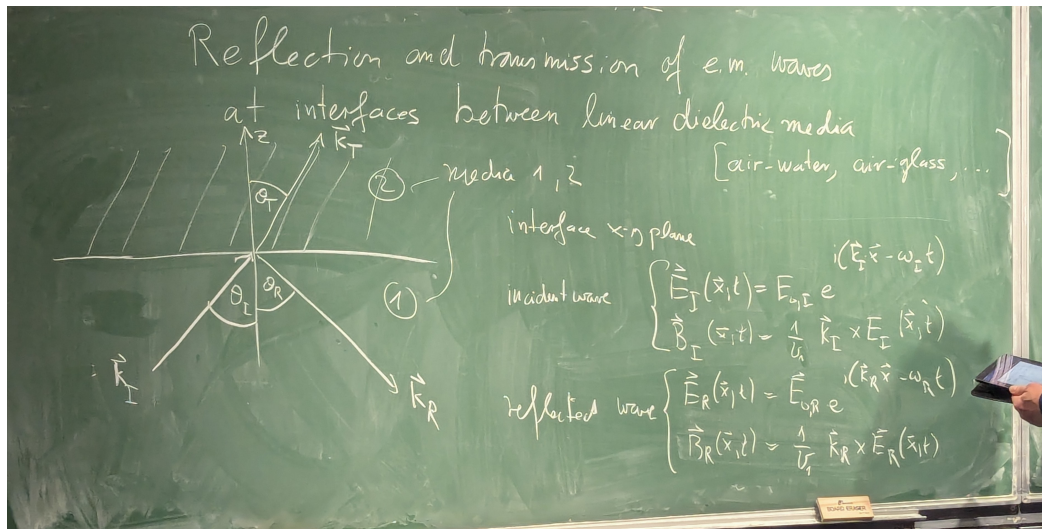


Figure 1: 8.1.jpg

While we have the transmitted wave to be

$$\begin{cases} \vec{E}_T(\vec{x}, t) = \vec{E}_{0T} e^{i(k_T \cdot \vec{x} - \omega_T t)} \\ \vec{B}_T(\vec{x}, t) = \frac{1}{v_2} \vec{k}_T \times \vec{E}_T \end{cases} \quad (22)$$

questions

- Relation between $\vec{k}_I, \vec{k}_R, \vec{k}_T$
- Relation between $\omega_I, \omega_R, \omega_T$
- size of reflected/transmitted amplitudes.
- same as the above one for intensities
- what about polarization?

Reflection and Transmission of EM wave at interfaces

Paul Haskins

April 2, 2025

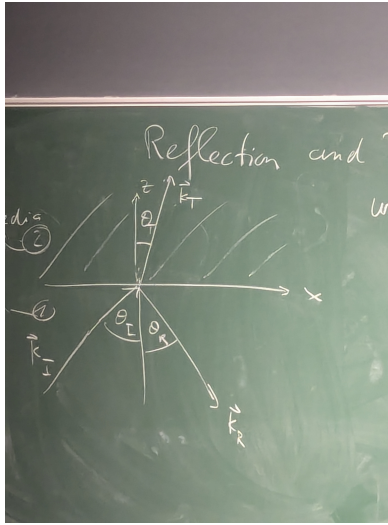


Figure 1:

We want to use the integral versions of Maxwell's equations

$$\int_S \vec{D} \cdot d\vec{S} = Q_{free, enclosed} = 0 \quad (1)$$

Vanishes only if dielectric. S is close surface that encloses the volume V.

$$\int_{\partial S} \vec{E} \cdot d\vec{r} = -\frac{d}{dt} \int_S \vec{B} \cdot d\vec{S} \quad (2)$$

where ∂S is the boundary of an open surface.

$$\int_{\partial S} \vec{H} \cdot d\vec{r} = I_{encl, free} + \frac{d}{dt} \int_S \vec{D} \cdot d\vec{S} \quad (3)$$

where $I_{enc} = 0$ for a dielectric.

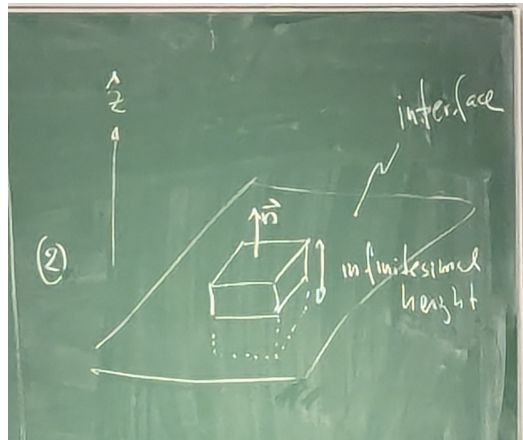


Figure 2:

Here the closed surface is the surface of the box. We are only concerned with up and down surfaces as the box has infinitesimal height. It faces 2 different media above and below.

$$\vec{n} \cdot (\vec{D}_2 - \vec{D}_1) = \sigma_{free} a = 0 \quad (4)$$

$$(5)$$

$$\vec{D}_1^\perp = \vec{D}_2^\perp \quad (6)$$

Analogous for $\vec{B}$

$$\vec{B}_1^\perp = \vec{B}_2^\perp \quad (7)$$

Similarly (Griffiths)

$$\vec{H}_1^\parallel = \vec{H}_2^\parallel \leftrightarrow \frac{1}{\mu_1} \vec{B}_1^\parallel = \frac{1}{\mu_2} \vec{B}_2^\parallel \quad (8)$$

ONLY HOLDS in linear media.
in our geometry $\parallel \hat{x}$, $\hat{y}$ and $\perp \hat{z}$

Summary

$$\epsilon_1 \vec{E}_1^\perp = \epsilon_2 \vec{E}_2^\perp \quad (9)$$

$$\vec{B}_1^\perp = \vec{B}_2^\perp \quad (10)$$

$$\vec{E}_1^\parallel = \vec{E}_2^\parallel \quad (11)$$

$$\frac{1}{\mu_1} \vec{B}_1^\parallel = \frac{1}{\mu_2} \vec{B}_2^\parallel \quad (12)$$

Consequences

$$\vec{E}_{0,I}^\parallel e^{i(\vec{k}\vec{x}-\omega_I t)} \Big|_{z=0} + \vec{E}_{0,R}^\parallel e^{i(\vec{k}\vec{x}-\omega_R t)} \Big|_{z=0} = \vec{E}_{0,T}^\parallel e^{i(\vec{k}\vec{x}-\omega_T t)} \Big|_{z=0} \quad (13)$$

and

$$\epsilon_1 \left(\vec{E}_{0,I}^\perp e^{i(\vec{k}\vec{x}-\omega_I t)} \Big|_{z=0} + \vec{E}_{0,R}^\perp e^{i(\vec{k}\vec{x}-\omega_R t)} \Big|_{z=0} \right) = \epsilon_2 \vec{E}_{0,T}^\perp e^{i(\vec{k}\vec{x}-\omega_T t)} \Big|_{z=0} \quad (14)$$

These must hold for any t and any $\vec{x} = (x, y, 0)^T$.

Not $\vec{x} = 0 \implies \omega_I = \omega_R = \omega_T = \omega$

$\rightarrow$ no change in frequency

If we set $t = 0$

$$\vec{k}_I \vec{x} \Big|_{z=0} = \vec{k}_R \vec{x} \Big|_{z=0} = \vec{k}_T \vec{x} \Big|_{z=0} \quad (15)$$

Choose incident wave vector in the $x - z$ plane

$$\vec{K}_I = k_I (\sin \theta_I \hat{x} + \cos \theta_I \hat{z}) \quad (16)$$

$$\vec{k}_R = k_R (\sin \theta_R \cos \phi_R \hat{x} + \sin \theta_R \sin \phi_R \hat{y} - \cos \theta_R \hat{z}) \quad (17)$$

$$\vec{k}_T = k_T (\sin \theta_R \cos \phi_R \hat{x} + \sin \theta_R \sin \phi_R \hat{y} + \cos \theta_R \hat{z}) \quad (18)$$

set

$$\vec{x} \Big|_{z=0} = \hat{x} \quad (19)$$

$$\vec{x} \cdot \vec{k}_I = k_I \sin \theta_I = k_R \sin \theta_R \cos \phi_R = k_T \sin \theta_T \cos \phi_T \quad (20)$$

set

$$\vec{x} \Big|_{z=0} = \hat{y} \quad (21)$$

$$0 = k_R \sin \theta_R \sin \phi_R = k_T \sin \theta_T \sin \phi_T \quad (22)$$

so

$$\phi_R = \phi_T = 0 \implies \vec{k}_I, \vec{k}_R, \vec{k}_T \quad (23)$$

are in $x - z$ plane. (Plane of incidence)
1st law of geometric optics

$$k_R \sin \theta_R = k_T \sin \theta_T = k_I \sin \theta_I \quad (24)$$